Analytical derivations

In[1]:=
```
ClearAll["Global\`*"]
```

1. Define species, complexes, and reaction directions

In[2]:=
```
(* Reaction-network stoichiometry, conservation laws, and deficiency *)

(*
  Companion to main-text Methods, "Structural network analysis" and
  Supplementary Note 1.

  Species order: A, AP, APP, K, P, AK, APK, APPC, APC.
  The 10 rows below are the complexes; edges encode the 12 directed
  reaction directions of the eight-step ordered distributive cycle.
*)
```

The network is encoded as ten complexes over nine species. Each directed edge is one reaction direction; reversible steps are represented by two directed edges.

In[3]:=
```
species = {A, AP, APP, K, P, AK, APK, APPC, APC}
```
Out[3]=
```
{A, AP, APP, K, P, AK, APK, APPC, APC}
```

In[4]:=
```
complexNames = {"A+K", "AK", "AP+K", "APK", "APP+K",
  "APP+P", "APPC", "AP+P", "APC", "A+P"}
```
Out[4]=
```
{A+K, AK, AP+K, APK, APP+K, APP+P, APPC, AP+P, APC, A+P}
```

In[5]:=
```
(* Complexes as rows of species-count vectors in the stated species order. *)
complexVectors = {
 {1, 0, 0, 1, 0, 0, 0, 0, 0}, (* A+K   *)
 {0, 0, 0, 0, 0, 1, 0, 0, 0}, (* AK    *)
 {0, 1, 0, 1, 0, 0, 0, 0, 0}, (* AP+K  *)
 {0, 0, 0, 0, 0, 0, 1, 0, 0}, (* APK   *)
 {0, 0, 1, 1, 0, 0, 0, 0, 0}, (* APP+K *)
 {0, 0, 1, 0, 1, 0, 0, 0, 0}, (* APP+P *)
 {0, 0, 0, 0, 0, 0, 0, 1, 0}, (* APPC  *)
 {0, 1, 0, 0, 1, 0, 0, 0, 0}, (* AP+P  *)
 {0, 0, 0, 0, 0, 0, 0, 0, 1}, (* APC   *)
 {1, 0, 0, 0, 1, 0, 0, 0, 0} (* A+P   *)
 }
```

Out[5]=
```
{{1, 0, 0, 1, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 1, 0, 0, 0},
 {0, 1, 0, 1, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 1, 0, 0},
 {0, 0, 1, 1, 0, 0, 0, 0, 0}, {0, 0, 1, 0, 1, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 1, 0},
 {0, 1, 0, 0, 1, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 1}, {1, 0, 0, 0, 1, 0, 0, 0, 0}}
```

In[6]:=
```
(* Each pair is {reactant-complex index, product-complex index}. *)
directedEdges = {
 {1, 2}, {2, 1}, {2, 3},
 {3, 4}, {4, 3}, {4, 5},
 {6, 7}, {7, 6}, {7, 8},
 {8, 9}, {9, 8}, {9, 10}
 }
```

Out[6]=
```
{{1, 2}, {2, 1}, {2, 3}, {3, 4}, {4, 3}, {4, 5}, {6, 7}, {7, 6}, {7, 8}, {8, 9}, {9, 8}, {9, 10}}
```

In[7]:=
```
(* Columns of N are product minus reactant stoichiometric vectors. *)
stoichiometricMatrix = Transpose[
 (complexVectors[[#[[2]]]] - complexVectors[[#[[1]]]]) & /@ directedEdges
 ]
```

Out[7]=
```
{{-1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1},
 {0, 0, 1, -1, 1, 0, 0, 0, 1, -1, 1, 0}, {0, 0, 0, 0, 0, 1, -1, 1, 0, 0, 0, 0},
 {-1, 1, 1, -1, 1, 1, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, -1, 1, 1, -1, 1, 1},
 {1, -1, -1, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 1, -1, -1, 0, 0, 0, 0, 0, 0},
 {0, 0, 0, 0, 0, 0, 1, -1, -1, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 0, 1, -1, -1}}
```

Added for the figures: set-up for the plot cells of this notebook. Evaluate it before any plot cell below. All names begin with "fig", so the derivations above and below are not affected.

In[8]:=
```
(* Shared set-up for the plot cells: evaluate this cell once, before any plot cell.
   All names start with "fig" so that they do not clash with the symbols of the
   derivation notebooks (k1, A, K, P, u, x, alpha, ... are left untouched).
   Every figure is shown and also exported as PDF to the sub-folder "plots"
   next to this notebook (or of the working directory if the notebook is unsaved). *)
figDir = Quiet[Check[FileNameJoin[{NotebookDirectory[], "plots"}], $Failed]];
If[! StringQ[figDir], figDir = FileNameJoin[{Directory[], "plots"}]];
```

```
If[! DirectoryQ[figDir], CreateDirectory[figDir]];
figSave[name_String, g_] := (Export[FileNameJoin[{figDir, name <> ".pdf"}], g]; g);
(* figShow prints every panel as its own output (at most 460 points wide, so that the no
   prints to PDF without scroll panes cutting the figures) and exports the composite fig
figShow[name_String, panels_List, composite_] := (
   Scan[Print, panels];
   Export[FileNameJoin[{figDir, name <> ".pdf"}], composite];
   Print[Style["saved: " <> FileNameJoin[{figDir, name <> ".pdf"}], Gray, 9]];);
(* number formatting without padding messages: n significant digits, integers when large
figNum[x_?NumericQ, n_Integer: 4] := StringReplace[
   If[Abs[x] >= 10^n, ToString[Round[x]], ToString[NumberForm[N[x], n]]], RegularExpression["\
figBlue = RGBColor[0.12, 0.38, 0.70]; figRed = RGBColor[0.84, 0.24, 0.24];

(* Parameter sets (concentrations in nM; k1, k3, k5, k7 in 1/(nM s); others in 1/s).
   figSet3: Iteration 1, row 96 (Class 5), the set of main-text Fig 3.
   figSet6: Iteration 3, row 5990 (Class 6), fully kinase-phosphatase symmetric. *)
figSet3 = <|"KTOT" -> 132.77894269306515, "PTOT" -> 16.907466299146936, "k1" -> 0.004470
figSet6 = <|"KTOT" -> 25.68678690797921, "PTOT" -> 25.68678690797921, "k1" -> 0.05546638

(* Constants of the inverse parametrisation, Eqs (7)-(9). *)
figConst[pr_Association] := Module[{m1, m3, m5, m7, a1, a2, r, al, be},
  m1 = (pr["k11"] + pr["k2"])/pr["k1"]; m3 = (pr["k33"] + pr["k4"])/pr["k3"];
  m5 = (pr["k55"] + pr["k6"])/pr["k5"]; m7 = (pr["k77"] + pr["k8"])/pr["k7"];
  a1 = pr["k2"] m7/(pr["k8"] m1); a2 = pr["k4"] m5/(pr["k6"] m3);
  r = pr["KTOT"]/pr["PTOT"];
  al = (a1/m3) (r pr["k4"]/pr["k6"] - 1); be = (1/m1) (r pr["k2"]/pr["k8"] - 1);
  <|"M1" -> m1, "M3" -> m3, "M5" -> m5, "M7" -> m7, "a1" -> a1, "a2" -> a2, "r" -> r,
    "alpha" -> al, "beta" -> be, "ustar" -> If[al != 0, -be/al, Infinity],
    "degenerate" -> (Abs[al r + be] <= 10^-9 (Abs[al] r + Abs[be]))|>];

(* Free substrate A(u), Eq. (8), and the full steady state at (u, A), Eq. (9). *)
figA[c_Association, u_] := (u - c["r"])/(u (c["alpha"] u + c["beta"]));
figState[pr_Association, c_Association, u_, a_] :=
 Module[{kk, pp, x1, x2, x3, x4, x5, x6},
  kk = pr["KTOT"]/(1 + a/c["M1"] + c["a1"] u a/c["M3"]); pp = kk/u;
  x1 = c["a1"] u a; x2 = c["a1"] c["a2"] u^2 a;
  x3 = a kk/c["M1"]; x4 = x1 kk/c["M3"]; x5 = x2 pp/c["M5"]; x6 = x1 pp/c["M7"];
  <|"u" -> u, "A" -> a, "K" -> kk, "P" -> pp, "x" -> {x1, x2, x3, x4, x5, x6},
    "ATOT" -> a + x1 + x2 + x3 + x4 + x5 + x6|>];

(* Reduced six-state model, Eqs (1)-(2); state order {AP, APP, AK, APK, APPC, APC}. *)
figRHS[pr_Association, at_, xs_List] := Module[{v, fA, fK, fP},
  fA = at - Total[xs]; fK = pr["KTOT"] - xs[[3]] - xs[[4]]; fP = pr["PTOT"] - xs[[5]] - xs[[6]];
  v = {pr["k1"] fA fK - pr["k11"] xs[[3]], pr["k2"] xs[[3]],
       pr["k3"] xs[[1]] fK - pr["k33"] xs[[4]], pr["k4"] xs[[4]],
       pr["k5"] xs[[2]] fP - pr["k55"] xs[[5]], pr["k6"] xs[[5]],
       pr["k7"] xs[[1]] fP - pr["k77"] xs[[6]], pr["k8"] xs[[6]]};
  {v[[2]] - v[[3]] + v[[6]] - v[[7]], v[[4]] - v[[5]], v[[1]] - v[[2]],
   v[[3]] - v[[4]], v[[5]] - v[[6]], v[[7]] - v[[8]]}];

(* Analytic Jacobian in the compact form of the notebook (support/jac6.m). *)
figJac[pr_Association, at_, xs_List] := Module[{x1, x2, x3, x4, x5, x6, s, cK, cP},
  {x1, x2, x3, x4, x5, x6} = xs; s = Total[xs] - at;
```

```
  cK = x3 + x4 - pr["KTOT"]; cP = x5 + x6 - pr["PTOT"];
  With[{k1 = pr["k1"], k11 = pr["k11"], k2 = pr["k2"], k3 = pr["k3"], k33 = pr["k33"],
        k4 = pr["k4"], k5 = pr["k5"], k55 = pr["k55"], k6 = pr["k6"], k7 = pr["k7"],
        k77 = pr["k77"], k8 = pr["k8"]},
   {{k3 cK + k7 cP, 0, k2 + x1 k3, k33 + x1 k3, k6 + x1 k7, k77 + x1 k7},
    {0, k5 cP, 0, k4, k55 + x2 k5, x2 k5},
    {k1 cK, k1 cK, k1 s - k11 - k2 + k1 cK, k1 s + k1 cK, k1 cK, k1 cK},
    {-k3 cK, 0, -x1 k3, -k4 - k33 - x1 k3, 0, 0},
    {0, -k5 cP, 0, 0, -k6 - k55 - x2 k5, -x2 k5},
    {-k7 cP, 0, 0, 0, -x1 k7, -k8 - k77 - x1 k7}}]];
figStableQ[pr_Association, at_, xs_List] := Max[Re[Eigenvalues[N[figJac[pr, at, xs]]]]] < 0;

(* All positive steady states as branches of the parametrisation. *)
figPosQ[s_Association] := Min[s["x"]] > 0 && s["K"] > 0 && s["P"] > 0;
figUGrid[c_Association] := Module[{crit = Select[{c["r"], c["ustar"]}, 0 < # < Infinity &]},
  Union[10.^Range[-14., 10., 0.004],
   Flatten[Table[cc {1 - 10.^-k, 1 + 10.^-k}, {cc, crit}, {k, 1, 12}]]]];
figBranches[pr_Association] := Module[{c = figConst[pr], r, crit, us, pts, br},
  r = c["r"];
  If[TrueQ[c["degenerate"]],
   (* symmetric case: the line u = r (any A) and the curve A = 1/(alpha u) *)
   br = {figState[pr, c, r, #] & /@ (10.^Range[-8., 8., 0.004])};
   If[c["alpha"] > 0,
    AppendTo[br, figState[pr, c, #, 1/(c["alpha"] #)] & /@ (10.^Range[-6., 6., 0.002])]],
   (* general case: the intervals of {u > 0 : A(u) > 0} between r and u* *)
   crit = Select[{r, c["ustar"]}, 0 < # < Infinity &];
   us = Select[figUGrid[c], figA[c, #] > 0 &];
   pts = figState[pr, c, #, figA[c, #]] & /@ us;
   br = GatherBy[pts, Function[s, Count[crit, cc_ /; cc < s["u"]]]]];
  br = Function[b, Select[b, figPosQ]] /@ br;
  Select[br, Length[#] > 1 &]];

(* Folds: extrema of ATOT along a branch; branch point of the symmetric case. *)
figFoldsOf[branch_List] := Module[{g = #["ATOT"] & /@ branch, d},
  d = Differences[Sign[Differences[g]]];
  branch[[1 + Flatten[Position[d, z_ /; z != 0, {1}]]]]];
figBP[pr_Association] := With[{c = figConst[pr]},
  If[TrueQ[c["degenerate"]] && c["alpha"] > 0,
   figState[pr, c, c["r"], 1/c["alpha"]], Missing["NoBranchPoint"]]];
figSingular[pr_Association, br_List] :=
  Join[Flatten[figFoldsOf /@ br, 1], If[MissingQ[figBP[pr]], {}, {figBP[pr]}]];

(* Exact bifurcation diagram APP against ATOT: blue stable, red unstable, black dots
   folds and branch points. Options (PlotRange, PlotLabel, ...) are passed on. *)
figDiagram[pr_Association, opts___] := Module[{br = figBranches[pr], lab, st, un, sing, da
  lab = Flatten[Function[b, {#["ATOT"], #["x"][[2]], figStableQ[pr, #["ATOT"], #["x"]]} & /@
  st = Cases[lab, {a_, b_, True} :> {a, b}]; un = Cases[lab, {a_, b_, False} :> {a, b}];
  sing = {#["ATOT"], #["x"][[2]]} & /@ figSingular[pr, br];
  data = Select[{{st, Directive[figBlue, PointSize[0.006]]}, {un, Directive[figRed, PointSi
      {sing, Directive[Black, PointSize[0.02]]}}, Length[First[#]] > 0 &];
  ListLogLogPlot[data[[All, 1]], opts, PlotStyle -> data[[All, 2]],
   PlotRange -> {{10^-1, 10^6}, All}, Frame -> True, FrameLabel -> {"A_TOT (nM)", "APP (nM)
   ImageSize -> 440]];
```

```
(* Steady states at a given ATOT (roots of ATOT(u) = at on every branch). *)
figStatesAt[pr_Association, at_] := Module[{c = figConst[pr], br, out = {}, g, lu, s},
  br = figBranches[pr];
  Do[g = #["ATOT"] - at & /@ b;
   Do[If[g[[j]] g[[j + 1]] <= 0,
     AppendTo[out,
      If[TrueQ[c["degenerate"]], b[[j]],
       s = FindRoot[figState[pr, c, 10^lu, figA[c, 10^lu]]["ATOT"] == at,
         {lu, Log10[b[[j]]["u"]], Log10[b[[j + 1]]["u"]]}];
       figState[pr, c, 10^(lu /. s), figA[c, 10^(lu /. s)]]]]],
    {j, Length[g] - 1}], {b, br}];
  SortBy[out, #["u"] &]];

(* Fold of the detached branch (minimum of ATOT(u) on the branch that does not reach u =
figDetachedFold[pr_Association] := Module[{c = figConst[pr], br, det, i0, lu, sol},
  If[TrueQ[c["degenerate"]], Return[Missing["NoDetachedBranch"]]];
  br = figBranches[pr];
  det = Select[br, Min[Abs[Log10[#["u"]/c["r"]]] & /@ #] > 10^-3 &];
  If[det === {}, Return[Missing["NoDetachedBranch"]]];
  det = First[det];
  i0 = First[Ordering[#["ATOT"] & /@ det, 1]];
  (* derivative-free search started from two points (no line search, no lstol message) *)
  sol = Quiet[FindMinimum[figState[pr, c, 10^lu, figA[c, 10^lu]]["ATOT"],
     {lu, Log10[det[[i0]]["u"]] - 0.01, Log10[det[[i0]]["u"]] + 0.01}], {FindMinimum::lstol}];
  figState[pr, c, 10^(lu /. sol[[2]]), figA[c, 10^(lu /. sol[[2]])]]];

(* Second derivatives and dF/dATOT in augmented coordinates (x1..x6, ATOT), and the
   right/left null vectors q, p of a singular Jacobian (|q| = 1, p.q = 1). *)
figHess[pr_Association, at_, xs_List] := Module[{z = Array[figZ, 7], f},
  f = figRHS[pr, z[[7]], z[[1 ;; 6]]];
  D[f, {z, 2}] /. Thread[z -> Append[xs, at]]];
figFA[pr_Association, at_, xs_List] := Module[{z7}, D[figRHS[pr, z7, xs], z7] /. z7 -> at];
figNull[jm_] := Module[{uu, ww, vv, q, p},
  {uu, ww, vv} = SingularValueDecomposition[N[jm]];
  q = vv[[All, -1]]; p = uu[[All, -1]]; q = q/Norm[q]; p = p/(p . q);
  {q, p, ww[[-1, -1]]/ww[[1, 1]]}];

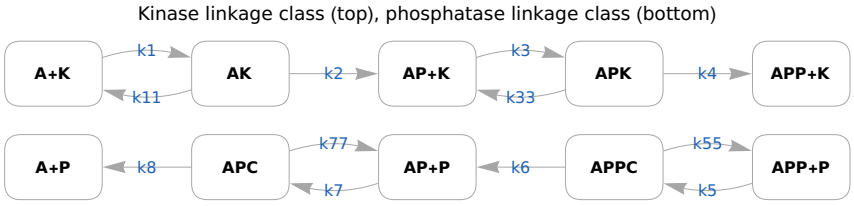
(* Branch point: exact branches near the branch point in linear axes, with the two
   tangent directions given by the branching equation a xi^2 + 2 b xi eta + c eta^2 = 0
figBPPlot[pr_Association, title_] := Module[{bp, xs, at, app, jm, q, p, fa, mm, sol, v0,
   a, b, cc, dirs, lines, br, lab, st, un, tmax},
  bp = figBP[pr]; If[MissingQ[bp], Return[Missing["NoBranchPoint"]]];
  xs = bp["x"]; at = bp["ATOT"]; app = xs[[2]];
  jm = figJac[pr, at, xs]; {q, p} = Most[figNull[jm]]; fa = figFA[pr, at, xs];
  mm = ArrayFlatten[{{jm, Transpose[{p}]}, {{q}, 0}}];
  sol = LinearSolve[mm, Append[-fa, 0]]; v0 = Most[sol];
  h = figHess[pr, at, xs]; qb = Append[q, 0]; vb = Append[v0, 1];
  a = p . (qb . # . qb & /@ h); b = p . (qb . # . vb & /@ h); cc = p . (vb . # . vb & /@
  dirs = If[Abs[a] < 10^-4 Max[Abs[{b, cc}]],
    {{1, 0}, {-cc/(2 b), 1}},
    {{(-b + Sqrt[b^2 - a cc])/a, 1}, {(-b - Sqrt[b^2 - a cc])/a, 1}}];
  lines = Function[d, With[{w = d[[1]] qb + d[[2]] vb}, {w[[7]], w[[2]]}] /@ dirs;
```

```
br = figBranches[pr];
lab = Flatten[Function[bb, {#["ATOT"], #["x"][[2]], figStableQ[pr, #["ATOT"], #["x"]]} & /@
lab = Select[lab, 0.4 at < #[[1]] < 1.6 at && #[[2]] < 3 app &];
st = Cases[lab, {x_, y_, True} :> {x, y}]; un = Cases[lab, {x_, y_, False} :> {x, y}];
Show[ListPlot[Select[{st, un}, Length[#] > 0 &],
  PlotStyle -> {Directive[figBlue, PointSize[0.006]], Directive[figRed, PointSize[0.006]]
 Graphics[{Black, Dashed, Thickness[0.004],
   Function[l, tmax = Min[If[Abs[l[[1]]] > 0, 0.5 at/Abs[l[[1]]], Infinity],
       If[Abs[l[[2]]] > 0, 1.2 app/Abs[l[[2]]], Infinity]];
     Line[{{at, app} - tmax l, {at, app} + tmax l}]] /@ lines,
   PointSize[0.025], Point[{at, app}]}],
 PlotRange -> {{0.4 at, 1.6 at}, {0, 3 app}}, Frame -> True,
 FrameLabel -> {"A_TOT (nM)", "APP (nM)"}, ImageSize -> 440,
 PlotLabel -> Style[Column[{title, Row[{"a = ", ScientificForm[a, 3], ",  b = ", Scienti
      ",  c = ", ScientificForm[cc, 3]}], Row[{"discriminant b^2 - a c = ", ScientificFo
      "  (dashed: tangents)"}]}, Alignment -> Center], 10]];
```

Added figure. Plot: the reaction network drawn from the complexes and the 12 directed reactions of Section 1 (two linkage classes, kinase above and phosphatase below). Saved as S1_reaction_network.pdf.

In[36]:=

```
(* Plot for Section 1: reaction network (10 complexes, 12 reaction directions, 2 linkage
figComplexNames = {"A+K", "AK", "AP+K", "APK", "APP+K", "APP+P", "APPC", "AP+P", "APC",
figEdges = {{1, 2}, {2, 1}, {2, 3}, {3, 4}, {4, 3}, {4, 5}, {6, 7}, {7, 6}, {7, 8}, {8, 9}, {9,
figEdgeLabels = {"k1", "k11", "k2", "k3", "k33", "k4", "k5", "k55", "k6", "k7", "k77", "
figNetwork = Graph[Range[10],
   MapThread[Labeled[DirectedEdge @@ #1, #2] &, {figEdges, figEdgeLabels}],
   VertexCoordinates -> Join[Table[{2 i, 1}, {i, 0, 4}], Table[{8 - 2 i, 0}, {i, 0, 4}]],
   VertexLabels -> Thread[Range[10] -> (Placed[Style[#, 9, Bold], Center] & /@ figComplexN
   VertexShapeFunction -> "RoundedRectangle", VertexSize -> {"Scaled", 0.11},
   VertexStyle -> Directive[White, EdgeForm[Gray]], EdgeStyle -> Gray,
   EdgeLabelStyle -> Directive[9, figBlue], ImageSize -> 460,
   PlotLabel -> Style["Kinase linkage class (top), phosphatase linkage class (bottom)", 1
figShow["S1_reaction_network", {figNetwork}, figNetwork]
```




```
saved:
  /wolframcloud/userfiles/9ff/9ffe219c-88d9-4035-be0c-26993d670000/Files/Harika/Derivation_Notebooks/Final/plots/S1_reaction_network.
  pdf
```

2. Verify conservation laws and deficiency

The three conserved totals must annihilate every stoichiometric reaction vector. The deficiency is c – l – rank(N), with ten complexes and two linkage classes.

In[41]:=
```
(* Totals written as coefficient vectors over the nine species. *)
substrateTotalVector = {1, 1, 1, 0, 0, 1, 1, 1, 1}
kinaseTotalVector = {0, 0, 0, 1, 0, 1, 1, 0, 0}
phosphataseTotalVector = {0, 0, 0, 0, 1, 0, 0, 1, 1}
conservationVectors = {substrateTotalVector, kinaseTotalVector,
 phosphataseTotalVector}
```

Out[41]=
```
{1, 1, 1, 0, 0, 1, 1, 1, 1}
```

Out[42]=
```
{0, 0, 0, 1, 0, 1, 1, 0, 0}
```

Out[43]=
```
{0, 0, 0, 0, 1, 0, 0, 1, 1}
```

Out[44]=
```
{{1, 1, 1, 0, 0, 1, 1, 1, 1}, {0, 0, 0, 1, 0, 1, 1, 0, 0}, {0, 0, 0, 0, 1, 0, 0, 1, 1}}
```

In[45]:=
```
(* Every conservation vector must annihilate every reaction direction. *)
conservationChecks = FullSimplify[Transpose[stoichiometricMatrix].#] & /@
 conservationVectors
```

Out[45]=
```
{{0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0},
 {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}}
```

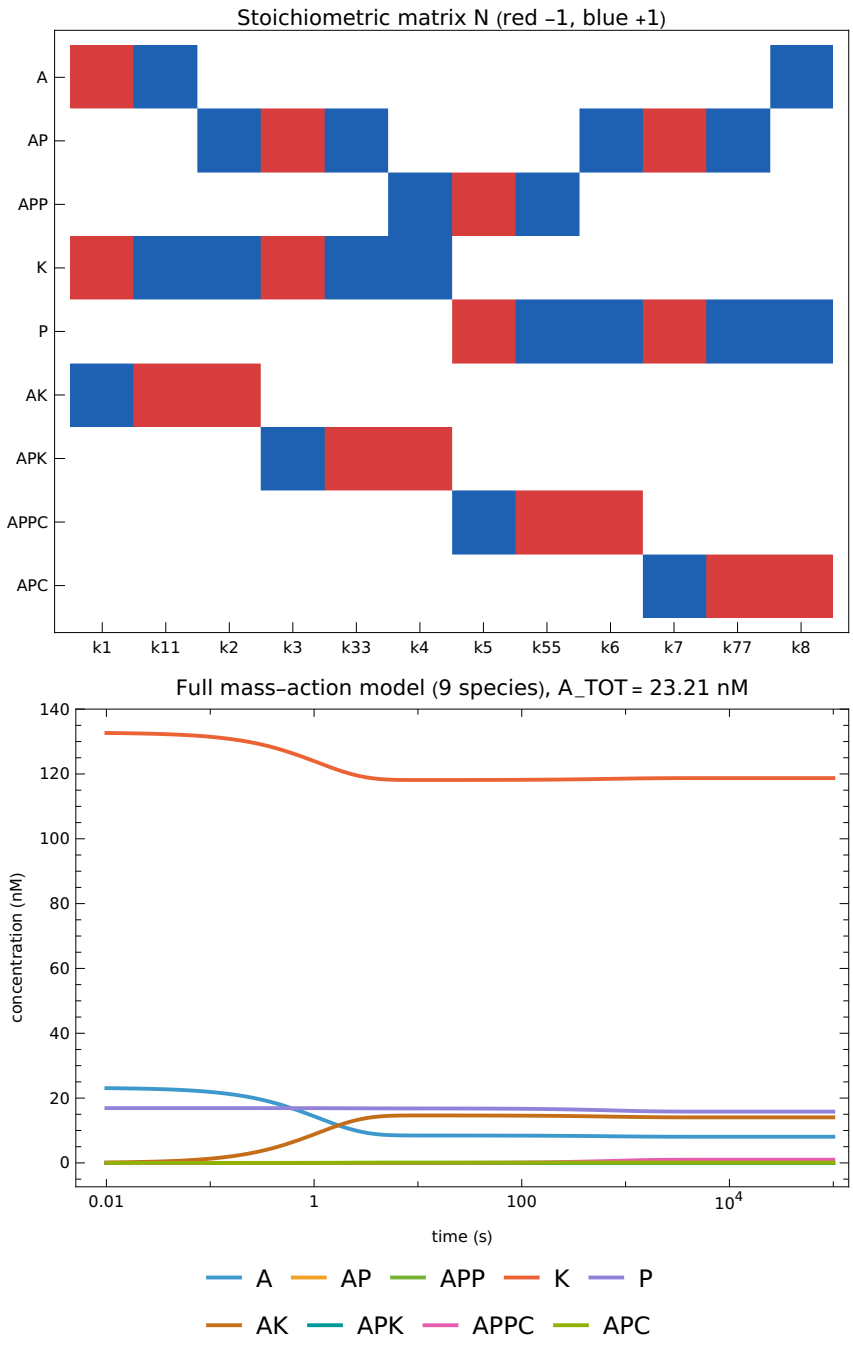
In[46]:=
```
complexCount = Length[complexVectors];
linkageClassCount = 2;
stoichiometricRank = MatrixRank[stoichiometricMatrix];
conservationDimension = Length[NullSpace[Transpose[stoichiometricMatrix]]];
networkDeficiency = complexCount - linkageClassCount - stoichiometricRank;
```

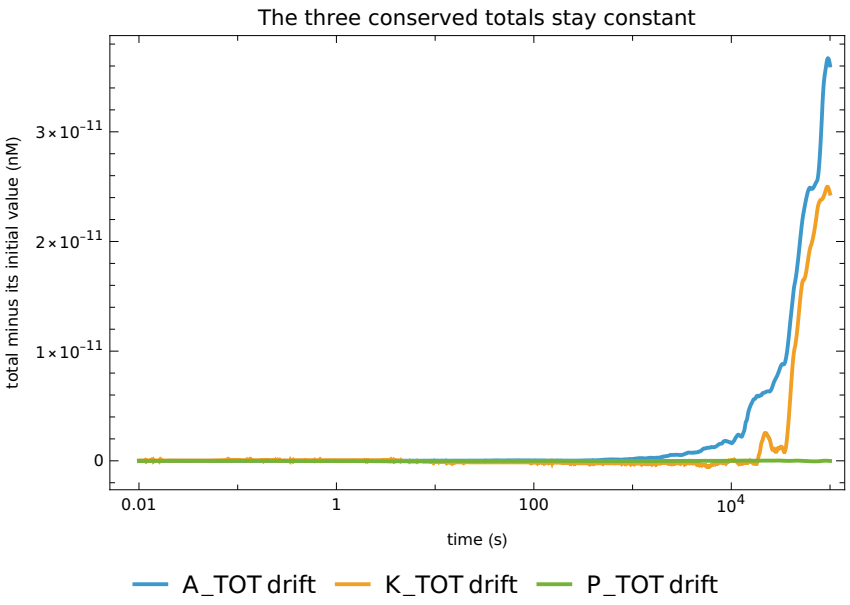
In[51]:=
```
Print["Species: ", Length[species], "; directed reaction vectors: ",
 Length[directedEdges]];
Print["Stoichiometric rank: ", stoichiometricRank,
 "; independent conservation laws: ", conservationDimension];
Print["Conservation identities (each should be a zero vector): ",
 conservationChecks];
Print["Deficiency c - l - rank(N): ", complexCount, " - ",
 linkageClassCount, " - ", stoichiometricRank, " = ", networkDeficiency];
```

```
Species: 9; directed reaction vectors: 12
Stoichiometric rank: 6; independent conservation laws: 3
Conservation identities (each should be a zero vector):
 {{0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0},
  {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}}
Deficiency c - l - rank(N): 10 - 2 - 6 = 2
```

Added figure. Plots: (left) the stoichiometric matrix N; (middle) a simulation of the full nine-species mass-action model for the Fig 3 parameter set at A_TOT = 23.21 nM; (right) the three conserved totals during that simulation, which stay at their initial values to round-off. Saved as S2_stoichiometry_and_conservation.pdf.

In[55]:=
```
(* Plot for Section 2: stoichiometric matrix and conservation of A_TOT, K_TOT, P_TOT in
figSpecies = {"A", "AP", "APP", "K", "P", "AK", "APK", "APPC", "APC"};
figComplexVectors = {{1, 0, 0, 1, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 1, 0, 0, 0}, {0, 1, 0,
   {0, 0, 0, 0, 0, 0, 1, 0, 0}, {0, 0, 1, 1, 0, 0, 0, 0, 0}, {0, 0, 1, 0, 1, 0, 0, 0, 0},
   {0, 0, 0, 0, 0, 0, 0, 1, 0}, {0, 1, 0, 0, 1, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 1},
   {1, 0, 0, 0, 1, 0, 0, 0, 0}};
figEdges = {{1, 2}, {2, 1}, {2, 3}, {3, 4}, {4, 3}, {4, 5}, {6, 7}, {7, 6}, {7, 8}, {8, 9}, {9,
figEdgeLabels = {"k1", "k11", "k2", "k3", "k33", "k4", "k5", "k55", "k6", "k7", "k77", "
figN = Transpose[(figComplexVectors[[#[[2]]]] - figComplexVectors[[#[[1]]]]) & /@ figEdges];
(* mass-action rate of each directed reaction: rate constant times its reactant concentr
figFullRHS[pr_Association, cs_List] := figN . MapThread[
    pr[#2] (Times @@ Pick[cs, figComplexVectors[[#1[[1]]]], 1]) &, {figEdges, figEdgeLabels}
figNPlot = MatrixPlot[figN,
   ColorFunction -> (Blend[{figRed, White, figBlue}, (# + 1)/2] &), ColorFunctionScaling ->
   FrameTicks -> {{Transpose[{Range[9], figSpecies}], None}, {Transpose[{Range[12], figEdgeLa
   PlotLabel -> "Stoichiometric matrix N (red −1, blue +1)", ImageSize -> 440];
Module[{pr = figSet3, at = 23.21, vars, c0, sol, tot, g1, g2},
 vars = Table[figC[i][figT], {i, 9}];
 c0 = {at, 0, 0, pr["KTOT"], pr["PTOT"], 0, 0, 0, 0};
 sol = First[NDSolve[Join[Thread[D[vars, figT] == figFullRHS[pr, vars]], Thread[(vars /. figT
    vars, {figT, 0, 10^5}, MaxSteps -> Infinity]];
 tot = {{1, 1, 1, 0, 0, 1, 1, 1, 1}, {0, 0, 0, 1, 0, 1, 1, 0, 0}, {0, 0, 0, 0, 1, 0, 0, 1
 g1 = LogLinearPlot[Evaluate[vars /. sol], {figT, 10^-2, 10^5}, PlotRange -> All, Frame ->
   PlotLegends -> Placed[figSpecies, Below], FrameLabel -> {"time (s)", "concentration (nM
   PlotLabel -> "Full mass-action model (9 species), A_TOT = 23.21 nM", ImageSize -> 440
 g2 = LogLinearPlot[Evaluate[(tot . vars - {at, pr["KTOT"], pr["PTOT"]}) /. sol], {figT, 10^-
   PlotRange -> All, Frame -> True, PlotLegends -> Placed[{"A_TOT drift", "K_TOT drift",
   FrameLabel -> {"time (s)", "total minus its initial value (nM)"},
   PlotLabel -> "The three conserved totals stay constant", ImageSize -> 440];
 figShow["S2_stoichiometry_and_conservation",{figNPlot, g1, g2}, GraphicsRow[{figNPlot,
```


Stoichiometric matrix N (red –1, blue +1)
A
AP
APP
K
P
AK
APK
APPC
APC
k1
k11
k2
k3
k33
k4
k5
k55
k6
k7
k77
k8
Full mass–action model (9 species), A_TOT = 23.21 nM
concentration (nM)
140
120
100
80
60
40
20
0
0.01
1
100
10^4
time (s)
A
AP
APP
K
P
AK
APK
APPC
APC



saved:
/wolframcloud/userfiles/9ff/9ffe219c-88d9-4035-be0c-26993d670000/Files/Harika/Derivation_Notebooks/Final/plots/S2_stoichiometry_and_conservation.pdf

3. Build the reduced mass-action model and Jacobian

In[63]:=
```
(* Reduced six-state mass-action equations and analytic derivatives *)

(*
  Companion to main-text Eqs. (1)--(2) and Methods, "Dynamical equations
  and conservation laws" and "Analytic derivatives and normal-form
  coefficients". The state order is x={AP, APP, AK, APK, APPC, APC};
  free A, K, and P are eliminated using the exact conservation laws.
*)
```

Free A, K, and P are eliminated using the conserved totals. Differentiating the six-state vector field gives the analytic Jacobian used by the bifurcation calculations.

In[64]:=
```
x = {x1, x2, x3, x4, x5, x6};
atot = ATOT; ktot = KTOT; ptot = PTOT;
freeA = atot - Total[x]
freeK = ktot - x3 - x4
freeP = ptot - x5 - x6
```

Out[66]=
$$ATOT - x1 - x2 - x3 - x4 - x5 - x6$$

Out[67]=
$$KTOT - x3 - x4$$

Out[68]=
$$PTOT - x5 - x6$$

In[69]:=
```
v1 = k1 freeA freeK - k11 x3
v2 = k2 x3
v3 = k3 x1 freeK - k33 x4
v4 = k4 x4
v5 = k5 x2 freeP - k55 x5
v6 = k6 x5
v7 = k7 x1 freeP - k77 x6
v8 = k8 x6
```

Out[69]=
$$-k11\ x3 + k1\ (KTOT - x3 - x4)\ (ATOT - x1 - x2 - x3 - x4 - x5 - x6)$$

Out[70]=
$$k2\ x3$$

Out[71]=
$$k3\ x1\ (KTOT - x3 - x4) - k33\ x4$$

Out[72]=
$$k4\ x4$$

Out[73]=
$$-k55\ x5 + k5\ x2\ (PTOT - x5 - x6)$$

Out[74]=
$$k6\ x5$$

Out[75]=
$$k7\ x1\ (PTOT - x5 - x6) - k77\ x6$$

Out[76]=
$$k8\ x6$$

In[77]:=
```
rhs = {v2 - v3 + v6 - v7,
  v4 - v5,
  v1 - v2,
  v3 - v4,
  v5 - v6,
  v7 - v8}
```

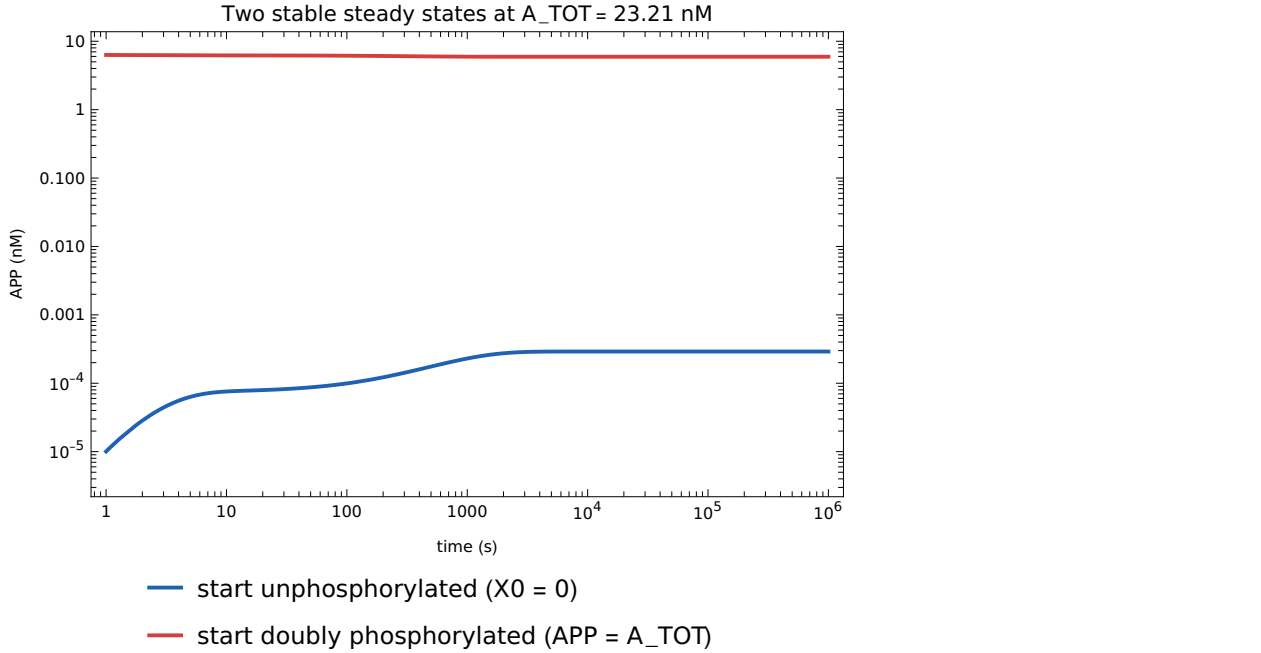
Out[77]=
$$\{k2\ x3 - k3\ x1\ (KTOT - x3 - x4) + k33\ x4 + k6\ x5 - k7\ x1\ (PTOT - x5 - x6) + k77\ x6,$$
$$k4\ x4 + k55\ x5 - k5\ x2\ (PTOT - x5 - x6),$$
$$-k11\ x3 - k2\ x3 + k1\ (KTOT - x3 - x4)\ (ATOT - x1 - x2 - x3 - x4 - x5 - x6),$$
$$k3\ x1\ (KTOT - x3 - x4) - k33\ x4 - k4\ x4,$$
$$-k55\ x5 - k6\ x5 + k5\ x2\ (PTOT - x5 - x6),\ k7\ x1\ (PTOT - x5 - x6) - k77\ x6 - k8\ x6\}$$

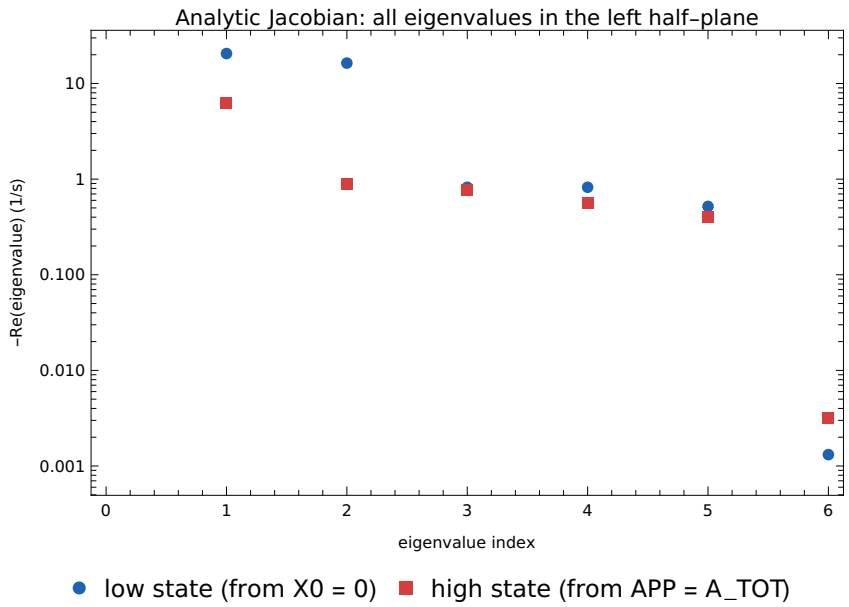
In[78]:=
```
(* Differentiate the right-hand side with respect to each state. *)
analyticJacobian = Table[D[rhs[[i]], x[[j]]], {i, 6}, {j, 6}]
```

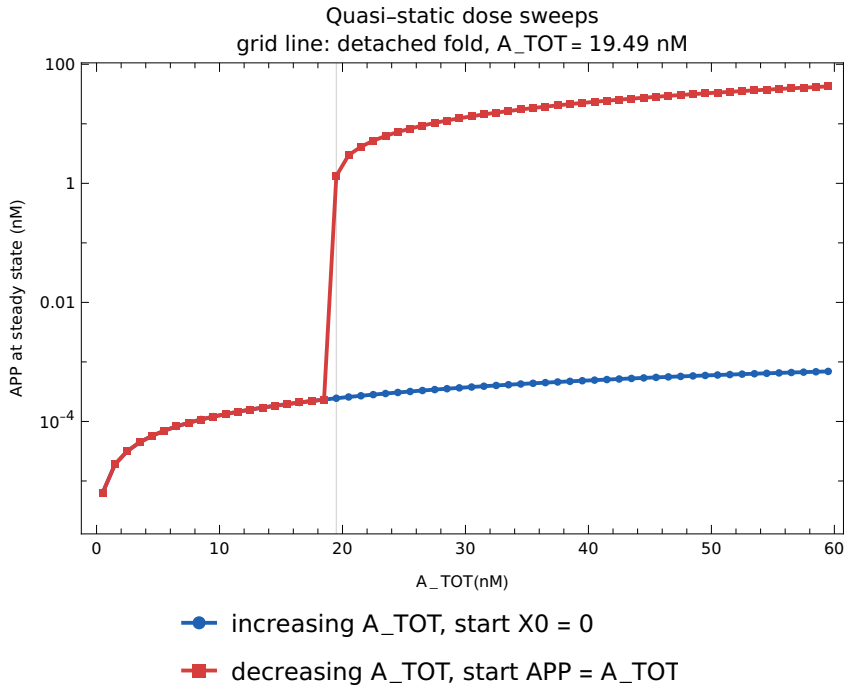
Out[78]=
```
{{-k3 (KTOT - x3 - x4) - k7 (PTOT - x5 - x6), 0, k2 + k3 x1, k33 + k3 x1, k6 + k7 x1, k77 + k7 x1},
 {0, -k5 (PTOT - x5 - x6), 0, k4, k55 + k5 x2, k5 x2}, {-k1 (KTOT - x3 - x4),
  -k1 (KTOT - x3 - x4), -k11 - k2 - k1 (KTOT - x3 - x4) - k1 (ATOT - x1 - x2 - x3 - x4 - x5 - x6),
  -k1 (KTOT - x3 - x4) - k1 (ATOT - x1 - x2 - x3 - x4 - x5 - x6), -k1 (KTOT - x3 - x4),
  -k1 (KTOT - x3 - x4)}, {k3 (KTOT - x3 - x4), 0, -k3 x1, -k33 - k4 - k3 x1, 0, 0},
 {0, k5 (PTOT - x5 - x6), 0, 0, -k55 - k6 - k5 x2, -k5 x2},
 {k7 (PTOT - x5 - x6), 0, 0, 0, -k7 x1, -k77 - k8 - k7 x1}}
```

Added figure. Plots for the Fig 3 parameter set: (left) APP(t) of the reduced six-state model at A_TOT = 23.21 nM from an unphosphorylated start (X0 = 0) and from a doubly phosphorylated start (APP = A_TOT): the two runs end in different stable steady states; (middle) minus the real parts of the eigenvalues of the analytic Jacobian at the two end states (all positive, so both states are stable); (right) a quasi-static dose sweep: A_TOT increased step by step from X0 = 0, and decreased step by step from APP = A_TOT, each step integrated to steady state. The increasing sweep stays on the main branch; the decreasing sweep stays on the detached branch until its fold (A_TOT = 19.49 nM) and then drops. Saved as S3_time_courses_eigenvalues_dose_sweep.pdf.

In[79]:=
```
(* Plot for Section 3: dynamics of the reduced model, Jacobian eigenvalues, dose-up and
Module[{pr = figSet3, at = 23.21, vars, run, xEnd, s0, s1, e0, e1, g1, g2, g3, grid, xu,
   upList = {}, downList = {}, fold},
 vars = Table[figY[i][figT], {i, 6}];
 run[x0_List, at1_, tmax_] := First[NDSolve[Join[Thread[D[vars, figT] == figRHS[pr, at1, va
     Thread[(vars /. figT -> 0) == x0]], vars, {figT, 0, tmax}, MaxSteps -> Infinity]];
 xEnd[s_, tmax_] := (vars /. s) /. figT -> tmax;
 s0 = run[ConstantArray[0., 6], at, 10^6]; s1 = run[{0., at, 0., 0., 0., 0.}, at, 10^6];
 g1 = LogLogPlot[Evaluate[{vars[[2]] /. s0, vars[[2]] /. s1}], {figT, 1, 10^6}, PlotRange -> A
   Frame -> True, PlotStyle -> {figBlue, figRed},
   PlotLegends -> Placed[{"start unphosphorylated (X0 = 0)", "start doubly phosphorylated
   FrameLabel -> {"time (s)", "APP (nM)"}, PlotLabel -> "Two stable steady states at A_TO
 e0 = Eigenvalues[figJac[pr, at, xEnd[s0, 10^6]]]; e1 = Eigenvalues[figJac[pr, at, xEnd[s1,
 g2 = ListLogPlot[{-Re[e0], -Re[e1]}, PlotMarkers -> {Automatic, 10}, PlotStyle -> {figBlue
   Frame -> True, FrameLabel -> {"eigenvalue index", "-Re(eigenvalue) (1/s)"},
   PlotLegends -> Placed[{"low state (from X0 = 0)", "high state (from APP = A_TOT)"}, Bel
   PlotLabel -> "Analytic Jacobian: all eigenvalues in the left half-plane", ImageSize
 grid = Range[0.5, 60., 1.];
 xu = ConstantArray[0., 6];
 Do[xu = xEnd[run[xu, a, 2 10^5], 2 10^5]; AppendTo[upList, {a, xu[[2]]}], {a, grid}];
 xd = {0., Last[grid], 0., 0., 0., 0.}; prev = Last[grid];
 Do[xd = xd a/prev; prev = a; xd = xEnd[run[xd, a, 2 10^5], 2 10^5]; AppendTo[downList, {a,
  {a, Reverse[grid]}];
 fold = figDetachedFold[pr]["ATOT"];
 g3 = ListLogPlot[{upList, downList}, Joined -> True, PlotMarkers -> {Automatic, 7},
   PlotStyle -> {figBlue, figRed}, Frame -> True, GridLines -> {{fold}, None},
   PlotLegends -> Placed[{"increasing A_TOT, start X0 = 0", "decreasing A_TOT, start APP
   FrameLabel -> {"A_TOT (nM)", "APP at steady state (nM)"},
   PlotLabel -> Column[{"Quasi-static dose sweeps", "grid line: detached fold, A_TOT = "
   ImageSize -> 440];
 figShow["S3_time_courses_eigenvalues_dose_sweep", {g1, g2, g3}, GraphicsRow[{g1, g2, g3},
```








```
saved:
  /wolframcloud/userfiles/9ff/9ffe219c-88d9-4035-be0c-26993d670000/Files/Harika/Derivation_Notebooks/Final/plots/S3_time_courses
  _eigenvalues_dose_sweep.pdf
```

4. Check the compact Jacobian formula

This cell compares the derivative computed directly from the reduced equations with the compact matrix used in the MATLAB implementation. A zero matrix means the two forms agree exactly.

In[80]:=
```
(* Compact form used by support/jac6.m in the MATLAB implementation. *)
s = Total[x] - atot; cK = x3 + x4 - ktot; cP = x5 + x6 - ptot;
compactJacobian = {
 {k3 cK + k7 cP, 0, k2 + x1 k3, k33 + x1 k3,
  k6 + x1 k7, k77 + x1 k7},
 {0, k5 cP, 0, k4, k55 + x2 k5, x2 k5},
 {k1 cK, k1 cK, k1 s - k11 - k2 + k1 cK,
  k1 s + k1 cK, k1 cK, k1 cK},
 {-k3 cK, 0, -x1 k3, -k4 - k33 - x1 k3, 0, 0},
 {0, -k5 cP, 0, 0, -k6 - k55 - x2 k5, -x2 k5},
 {-k7 cP, 0, 0, 0, -x1 k7, -k8 - k77 - x1 k7}
 }
```

Out[81]=
```
{{k3 (-KTOT + x3 + x4) + k7 (-PTOT + x5 + x6), 0, k2 + k3 x1, k33 + k3 x1, k6 + k7 x1, k77 + k7 x1},
 {0, k5 (-PTOT + x5 + x6), 0, k4, k55 + k5 x2, k5 x2}, {k1 (-KTOT + x3 + x4),
  k1 (-KTOT + x3 + x4), -k11 - k2 + k1 (-KTOT + x3 + x4) + k1 (-ATOT + x1 + x2 + x3 + x4 + x5 + x6),
  k1 (-KTOT + x3 + x4) + k1 (-ATOT + x1 + x2 + x3 + x4 + x5 + x6), k1 (-KTOT + x3 + x4),
  k1 (-KTOT + x3 + x4)}, {-k3 (-KTOT + x3 + x4), 0, -k3 x1, -k33 - k4 - k3 x1, 0, 0},
 {0, -k5 (-PTOT + x5 + x6), 0, 0, -k55 - k6 - k5 x2, -k5 x2},
 {-k7 (-PTOT + x5 + x6), 0, 0, 0, -k7 x1, -k77 - k8 - k7 x1}}
```

In[82]:=
```
jacobianResidual = FullSimplify[analyticJacobian - compactJacobian]
parameterDerivative = FullSimplify[D[rhs, atot]]
```

Out[82]=
```
{{0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0},
 {0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0}}
```

Out[83]=
```
{0, 0, k1 (KTOT - x3 - x4), 0, 0, 0}
```

In[84]:=
```
(* The vector field is quadratic in the states; its third state derivatives
   therefore vanish. The Hessian tensor is retained for file 06. *)
stateHessian = Table[
 D[rhs[[i]], x[[j]], x[[k]]], {i, 6}, {j, 6}, {k, 6}];
thirdStateDerivatives = Table[
 D[rhs[[i]], x[[j]], x[[k]], x[[ell]]],
 {i, 6}, {j, 6}, {k, 6}, {ell, 6}];
```

In[86]:=
```
Print["Reduced ODEs F: "]; Print[TraditionalForm /@ rhs];
Print["Analytic Jacobian agrees with the compact MATLAB Jacobian: ",
 jacobianResidual === ConstantArray[0, {6, 6}]];
Print["dF/dATOT: ", parameterDerivative];
Print["All third state derivatives vanish: ",
 Flatten[thirdStateDerivatives] === ConstantArray[0, 6^4]];
```

```
Reduced ODEs F:

{k2 x3 – k3 x1 (KTOT – x3 – x4) + k33 x4 + k6 x5 – k7 x1 (PTOT – x5 – x6) + k77 x6,
 k4 x4 – k5 x2 (PTOT – x5 – x6) + k55 x5,
 k1 (KTOT – x3 – x4) (ATOT – x1 – x2 – x3 – x4 – x5 – x6) – k11 x3 – k2 x3,
 k3 x1 (KTOT – x3 – x4) – k33 x4 – k4 x4,
 k5 x2 (PTOT – x5 – x6) – k55 x5 – k6 x5, k7 x1 (PTOT – x5 – x6) – k77 x6 – k8 x6}

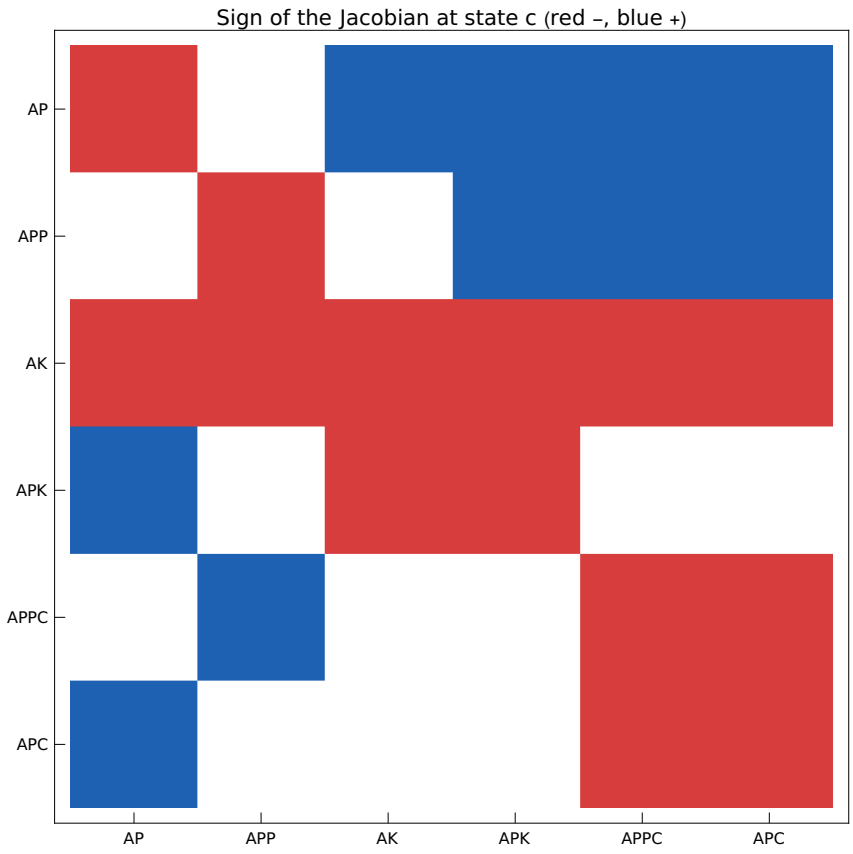
Analytic Jacobian agrees with the compact MATLAB Jacobian: True

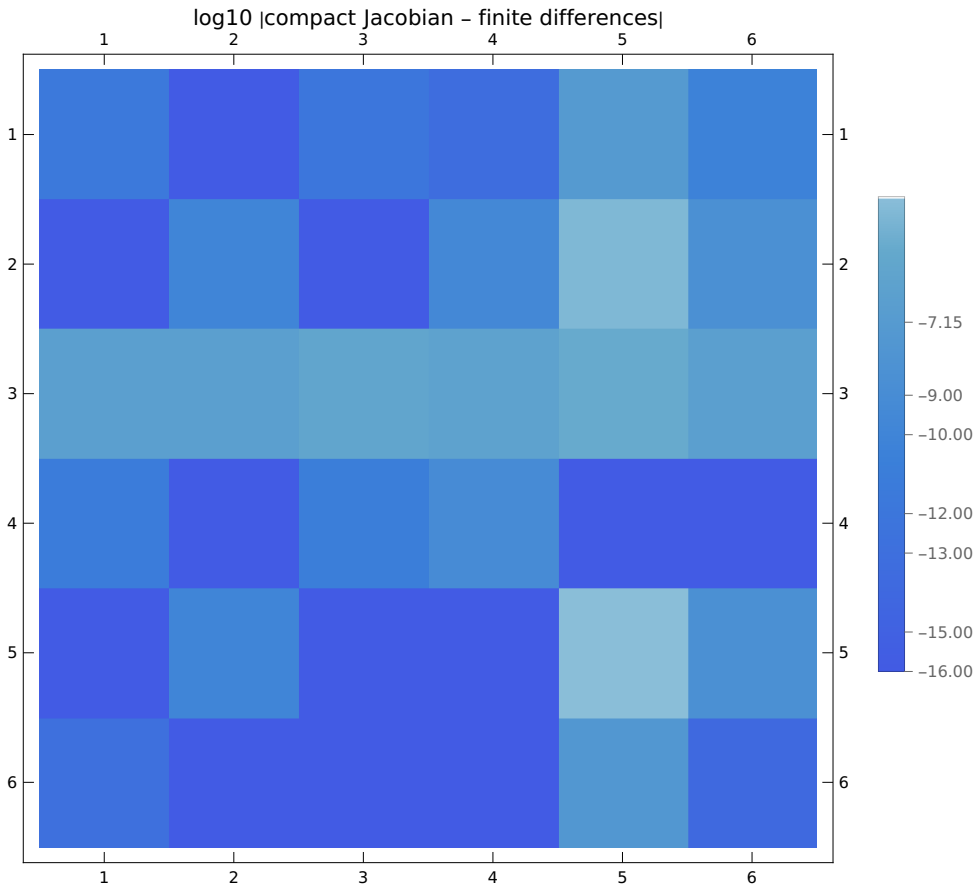
dF/dATOT: {0, 0, k1 (KTOT – x3 – x4), 0, 0, 0}

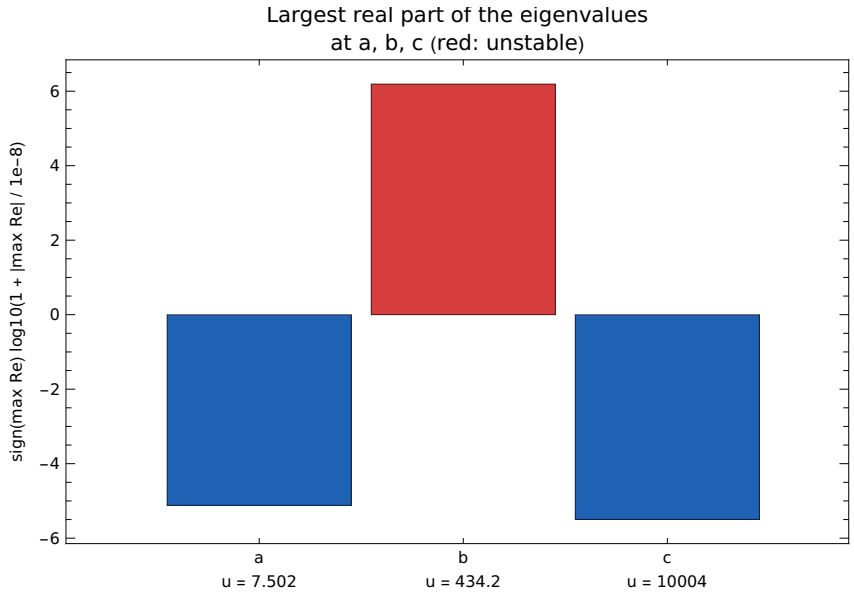
All third state derivatives vanish: True
```

Added figure. Plots at the three steady states a, b, c of the Fig 3 set at A_TOT = 23.21 nM: (left) sign pattern of the compact analytic Jacobian at state c; (middle) log10 of the entry-wise difference between the compact Jacobian and a central finite-difference Jacobian of the reduced model (round-off level); (right) the largest real part of the eigenvalues at a, b and c (b is the unstable middle state). Saved as S4_compact_Jacobian_checks.pdf.

In[90]:=
```
(* Plot for Section 4: compact Jacobian against finite differences, and stability of the
Module[{pr = figSet3, at = 23.21, sts, xs, jm, jfd, hh = 10^-7, g1, g2, g3, mx},
 sts = figStatesAt[pr, at]; xs = sts[[-1]]["x"];
 jm = figJac[pr, at, xs];
 jfd = Transpose[Table[(figRHS[pr, at, xs + hh UnitVector[6, j]] - figRHS[pr, at, xs - hh U
 g1 = MatrixPlot[Sign[jm], ColorFunction -> (Blend[{figRed, White, figBlue}, (# + 1)/2] &),
   FrameTicks -> {{Transpose[{Range[6], {"AP", "APP", "AK", "APK", "APPC", "APC"}}], None},
     {Transpose[{Range[6], {"AP", "APP", "AK", "APK", "APPC", "APC"}}], None}},
   PlotLabel -> "Sign of the Jacobian at state c (red –, blue +)", ImageSize -> 440];
 g2 = MatrixPlot[Log10[Abs[jm - jfd] + 10^-16], PlotLegends -> Automatic,
   PlotLabel -> "log10 |compact Jacobian - finite differences|", ImageSize -> 440];
 mx = Max[Re[Eigenvalues[figJac[pr, at, #["x"]]]]] & /@ sts;
 g3 = BarChart[Sign[mx] Log10[1 + Abs[mx]/10^-8], ChartLabels -> (Column[{#1, "u = " <> figN
   ChartStyle -> (If[# < 0, figBlue, figRed] & /@ mx), Frame -> True,
   FrameLabel -> {None, "sign(max Re) log10(1 + |max Re| / 1e-8)"},
   PlotLabel -> Column[{"Largest real part of the eigenvalues", "at a, b, c (red: unstab
 figShow["S4_compact_Jacobian_checks",{g1, g2, g3}, GraphicsRow[{g1, g2, g3}, ImageSize -
```


Sign of the Jacobian at state c (red –, blue +)
AP
APP
AK
APK
APPC
APC
AP
APP
AK
APK
APPC
APC





saved:
/wolframcloud/userfiles/9ff/9ffe219c-88d9-4035-be0c-26993d670000/Files/Harika/Derivation_Notebooks/Final/plots/S4_compact_Jacobian_checks.pdf

5. Derive the exact positive steady-state parametrisation

In[91]:=
```
(* Exact inverse parametrisation of every positive steady state *)

(*
  Companion to main-text Eqs. (7)--(9) and Methods, "Exact steady-state
  structure: the inverse method". This is an exact mass-action derivation;
  it does not invoke a quasi-steady-state approximation.
*)
```

The inverse method uses the ratio u = K/P and exact mass-action flux balances. It derives the four Michaelis constants, free substrate A(u), and the remaining concentrations without a quasi-steady-state approximation.

In[92]:=
```
(* Michaelis constants for the four enzyme-substrate complexes. *)
M1 = (k11 + k2)/k1;
M3 = (k33 + k4)/k3;
M5 = (k55 + k6)/k5;
M7 = (k77 + k8)/k7;
```

In[96]:=
```
(* The four complex steady-state equations. *)
complexSteadyRelations = {
 x3 → A K/M1, x4 → x1 K/M3,
 x5 → x2 P/M5, x6 → x1 P/M7}
```

Out[96]=

$$\left\{x3 \to \frac{A\ K\ k1}{k11 + k2},\ x4 \to \frac{K\ k3\ x1}{k33 + k4},\ x5 \to \frac{k5\ P\ x2}{k55 + k6},\ x6 \to \frac{k7\ P\ x1}{k77 + k8}\right\}$$

In[97]:=
```
(* Site-flux balances v2=v8 and v4=v6 give x1 and x2. *)
a1 = k2 M7/(k8 M1)
a2 = k4 M5/(k6 M3)
```

Out[97]=

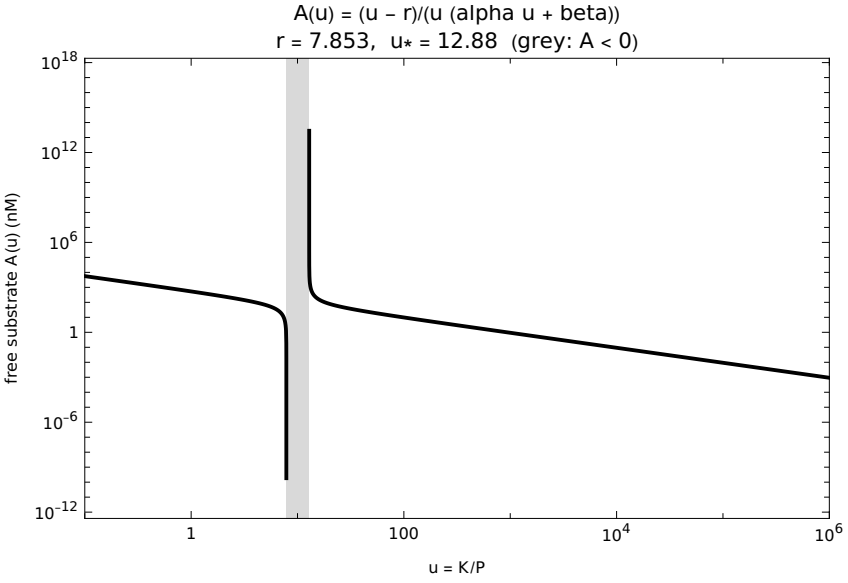
$$\frac{k1\ k2\ (k77 + k8)}{(k11 + k2)\ k7\ k8}$$

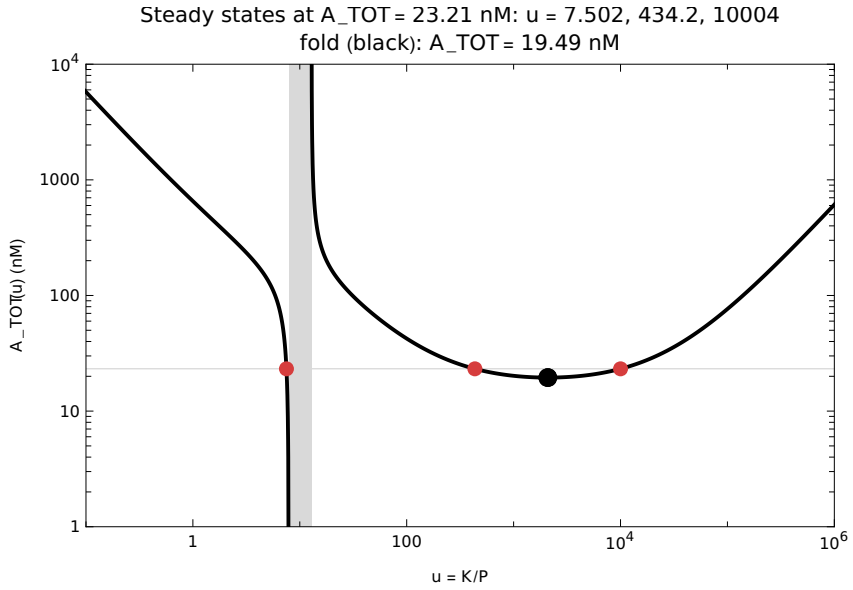
Out[98]=

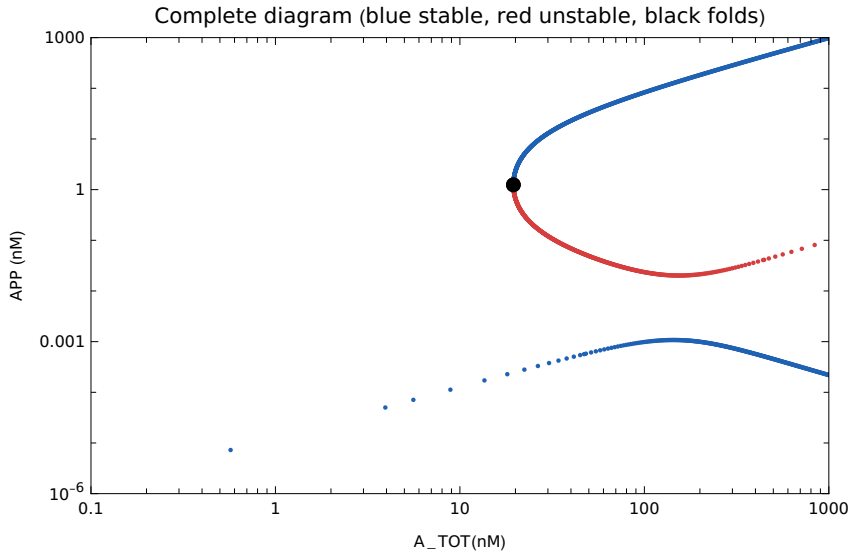
$$\frac{k3\ k4\ (k55 + k6)}{(k33 + k4)\ k5\ k6}$$

Added figure. Plots for the Fig 3 parameter set: (left) free substrate A(u) on its two positive intervals, with the band r < u < u* where A < 0 (grey); (middle) A_TOT(u): its level set at A_TOT = 23.21 nM gives the three steady states (open circles), its minimum on the detached interval is the fold (black); (right) the complete bifurcation diagram APP against A_TOT obtained by placing every grid point at its A_TOT (blue stable, red unstable). Saved as S5_parametrisation_A_ATOT_diagram.pdf.

In[99]:=
```
(* Plot for Section 5: A(u), ATOT(u) and the complete bifurcation diagram *)
Module[{pr = figSet3, c = figConst[figSet3], br, band, g1, g2, g3, roots, fold, curves},
 br = figBranches[pr];
 band = {GrayLevel[0.85], Rectangle[{Log[c["r"]], -100}, {Log[c["ustar"]], 100}]};
 curves = Function[b, {#["u"], #["A"]} & /@ b] /@ br;
 g1 = ListLogLogPlot[curves, Joined -> True, PlotStyle -> Black, Prolog -> band, Frame -
   PlotRange -> {{10^-1, 10^6}, All}, FrameLabel -> {"u = K/P", "free substrate A(u) (nM)"},
   PlotLabel -> Column[{"A(u) = (u - r)/(u (alpha u + beta))", "r = " <> figNum[c["r"]] <> ",
   ImageSize -> 440];
 roots = {#["u"], #["ATOT"]} & /@ figStatesAt[pr, 23.21];
 fold = With[{f = figDetachedFold[pr]}, {{f["u"], f["ATOT"]}}];
 curves = Function[b, {#["u"], #["ATOT"]} & /@ b] /@ br;
 g2 = ListLogLogPlot[Join[curves, {roots, fold}], Joined -> Join[ConstantArray[True, Lengt
   PlotStyle -> Join[ConstantArray[Black, Length[curves]], {Directive[figRed, PointSize[0.(
   Prolog -> band, Frame -> True, PlotRange -> {{10^-1, 10^6}, {1, 10^4}},
   GridLines -> {None, {23.21}}, FrameLabel -> {"u = K/P", "A_TOT(u)(nM)"},
   PlotLabel -> Column[{"Steady states at A_TOT = 23.21 nM: u = " <> StringRiffle[figNum
      "fold (black): A_TOT = " <> figNum[fold[[1, 2]]] <> " nM"}, Alignment -> Center], Imag
 g3 = figDiagram[pr, PlotRange -> {{10^-1, 10^3}, {10^-6, 10^3}},
   PlotLabel -> "Complete diagram (blue stable, red unstable, black folds)", ImageSize -
 figShow["S5_parametrisation_A_ATOT_diagram",{g1, g2, g3}, GraphicsRow[{g1, g2, g3}, Imag
```







saved:
/wolframcloud/userfiles/9ff/9ffe219c-88d9-4035-be0c-26993d670000/Files/Harika/Derivation_Notebooks/Final/plots/S5_parametrisation_A_ATOT_diagram.pdf

6. Check the flux-balance and total-balance identities

Substitution into the two site-flux balances should give zero residuals. The total-enzyme ratio equation should also reproduce A(u); these symbolic checks expose algebra errors before numerical use.

In[100]:=

```
(* u is kept as an independent symbolic parameter representing K/P. *)

x1FromFlux = Solve[
 k2 (A K / M1) == k8 (x1 P / M7), x1][[1, 1, 2]]
x1RatioResidual = FullSimplify[
 (x1FromFlux /. K → u P) - a1 u A]
```

Out[100]=

$$\frac{A\ K\ k1\ k2\ (k77 + k8)}{(k11 + k2)\ k7\ k8\ P}$$

Out[101]=

```
0
```

In[102]:=

```
x2FromFlux = Solve[
 k4 (x1 K / M3) == k6 (x2 P / M5), x2][[1, 1, 2]]
x2RatioResidual = FullSimplify[
 (x2FromFlux /. {K → u P, x1 → a1 u A}) - a1 a2 u^2 A]
```

Out[102]=

$$\frac{K\ k3\ k4\ (k55 + k6)\ x1}{(k33 + k4)\ k5\ k6\ P}$$

Out[103]=

```
0
```

In[104]:=

```
(* Divide the total-enzyme balances to eliminate the free-enzyme scale. *)
r = KT / PT;
ratioEquation = r == (u (1 + A / M1 + a1 u A / M3)) / (1 + a1 a2 u^2 A / M5 + a1 u A / M7)
```

Out[105]=

$$\frac{KT}{PT} == \frac{u\left(1 + \frac{A\ k1}{k11+k2} + \frac{A\ k1\ k2\ k3\ (k77+k8)\ u}{(k11+k2)\ (k33+k4)\ k7\ k8}\right)}{1 + \frac{A\ k1\ k2\ u}{(k11+k2)\ k8} + \frac{A\ k1\ k2\ k3\ k4\ (k77+k8)\ u^2}{(k11+k2)\ (k33+k4)\ k6\ k7\ k8}}$$

In[106]:=

```
alpha = (a1 / M3) (r k4 / k6 - 1)
beta = (1 / M1) (r k2 / k8 - 1)
```

Out[106]=

$$\frac{k1\ k2\ k3\ (k77 + k8)\left(-1 + \frac{k4\ KT}{k6\ PT}\right)}{(k11 + k2)\ (k33 + k4)\ k7\ k8}$$

Out[107]=

$$\frac{k1\left(-1 + \frac{k2\ KT}{k8\ PT}\right)}{k11 + k2}$$

In[108]:=
```
AofUFromTotals = FullSimplify[A /. First[Solve[ratioEquation, A]]]
```
Out[108]=
$$-\frac{(k11 + k2)\,(k33 + k4)\,k6\,k7\,k8\,(KT - PT\,u)}{k1\,u\,((k33 + k4)\,k6\,k7\,(k2\,KT - k8\,PT) + k2\,k3\,(k77 + k8)\,(k4\,KT - k6\,PT)\,u)}$$

In[109]:=
```
AofU = (u - r)/(u (alpha u + beta))
```
Out[109]=
$$\frac{-\frac{KT}{PT} + u}{u\left(\frac{k1\left(-1+\frac{k2\,KT}{k8\,PT}\right)}{k11+k2} + \frac{k1\,k2\,k3\,(k77+k8)\left(-1+\frac{k4\,KT}{k6\,PT}\right)u}{(k11+k2)\,(k33+k4)\,k7\,k8}\right)}$$

In[110]:=
```
parametrisationResidual = FullSimplify[
 Together[AofUFromTotals - AofU],
 Assumptions → {k1 > 0, k2 > 0, k3 > 0, k4 > 0, k5 > 0,
  k6 > 0, k7 > 0, k8 > 0, M1 > 0, M3 > 0, M5 > 0,
  M7 > 0, r > 0, u > 0}]
```
Out[110]=
```
0
```

In[111]:=
```
(* Reconstruct the free enzymes, all six states, and the substrate total. *)
KofU = KT/(1 + AofU/M1 + a1 u AofU/M3);
PofU = KofU/u;
x1ofU = a1 u AofU;
x2ofU = a1 a2 u^2 AofU;
x3ofU = AofU KofU/M1;
x4ofU = x1ofU KofU/M3;
x5ofU = x2ofU PofU/M5;
x6ofU = x1ofU PofU/M7;
ATOTofU = Together[AofU + x1ofU + x2ofU + x3ofU + x4ofU + x5ofU + x6ofU]
```
Out[119]=
$$-\Big(\big((KT - PT\,u)\,\big(-k11\,k33^2\,k5\,k6^2\,k7^2\,k8^2 - k2\,k33^2\,k5\,k6^2\,k7^2\,k8^2 - 2\,k11\,k33\,k4\,k5\,k6^2\,k7^2\,k8^2 - 2\,k2\,k33\,k4\,k5\,k6^2\,k7^2\,k8^2 - k11\,k4^2\,k5\,k6^2\,k7^2\,k8^2 - k2\,k4^2\,k5\,k6^2\,k7^2\,k8^2 + k11\,k2\,k33^2\,k5\,k6^2\,k7^2\,k8\,u + k2^2\,k33^2\,k5\,k6^2\,k7^2\,k8\,u + 2\,k11\,k2\,k33\,k4\,k5\,k6^2\,k7^2\,k8\,u + 2\,k2^2\,k33\,k4\,k5\,k6^2\,k7^2\,k8\,u + k11\,k2\,k4^2\,k5\,k6^2\,k7^2\,k8\,u + k2^2\,k4^2\,k5\,k6^2\,k7^2\,k8\,u - k11\,k2\,k3\,k33\,k5\,k6^2\,k7\,k77\,k8\,u - k2^2\,k3\,k33\,k5\,k6^2\,k7\,k77\,k8\,u - k1\,k2\,k33^2\,k5\,k6^2\,k7\,k77\,k8\,u - k11\,k2\,k3\,k4\,k5\,k6^2\,k7\,k77\,k8\,u - k2^2\,k3\,k4\,k5\,k6^2\,k7\,k77\,k8\,u - 2\,k1\,k2\,k33\,k4\,k5\,k6^2\,k7\,k77\,k8\,u - k1\,k2\,k4^2\,k5\,k6^2\,k7\,k77\,k8\,u - k11\,k2\,k3\,k33\,k5\,k6^2\,k7\,k8^2\,u - k2^2\,k3\,k33\,k5\,k6^2\,k7\,k8^2\,u - k1\,k2\,k33^2\,k5\,k6^2\,k7\,k8^2\,u - k11\,k2\,k3\,k4\,k5\,k6^2\,k7\,k8^2\,u - k2^2\,k3\,k4\,k5\,k6^2\,k7\,k8^2\,u - 2\,k1\,k2\,k33\,k4\,k5\,k6^2\,k7\,k8^2\,u - k1\,k2\,k4^2\,k5\,k6^2\,k7\,k8^2\,u + k1\,k2^2\,k33^2\,k5\,k6^2\,k7^2\,KT\,u + 2\,k1\,k2^2\,k33\,k4\,k5\,k6^2\,k7^2\,KT\,u + k1\,k2^2\,k4^2\,k5\,k6^2\,k7^2\,KT\,u + k1\,k2\,k33^2\,k5\,k6^2\,k7^2\,k8\,KT\,u + 2\,k1\,k2\,k33\,k4\,k5\,k6^2\,k7^2\,k8\,KT\,u + k1\,k2\,k4^2\,k5\,k6^2\,k7^2\,k8\,KT\,u - k1\,k2\,k33^2\,k5\,k6^2\,k7^2\,k8\,PT\,u - 2\,k1\,k2\,k33\,k4\,k5\,k6^2\,k7^2\,k8\,PT\,u - k1\,k2\,k4^2\,k5\,k6^2\,k7^2\,k8\,PT\,u - k1\,k33^2\,k5\,k6^2\,k7^2\,k8^2\,PT\,u - 2\,k1\,k33\,k4\,k5\,k6^2\,k7^2\,k8^2\,PT\,u -$$

$$
\begin{aligned}
&k1\,k4^2\,k5\,k6^2\,k7^2\,k8^2\,PT\,u + k1\,k2^2\,k33^2\,k5\,k6^2\,k7\,k77\,u^2 + 2\,k1\,k2^2\,k33\,k4\,k5\,k6^2\,k7\,k77\,u^2 + \\
&k1\,k2^2\,k4^2\,k5\,k6^2\,k7\,k77\,u^2 - k1\,k2^2\,k3\,k33\,k5\,k6^2\,k77^2\,u^2 - k1\,k2^2\,k3\,k4\,k5\,k6^2\,k77^2\,u^2 + \\
&k1\,k2^2\,k33^2\,k5\,k6^2\,k7\,k8\,u^2 + 2\,k1\,k2^2\,k33\,k4\,k5\,k6^2\,k7\,k8\,u^2 + k1\,k2^2\,k4^2\,k5\,k6^2\,k7\,k8\,u^2 - \\
&2\,k1\,k2^2\,k3\,k33\,k5\,k6^2\,k77\,k8\,u^2 - 2\,k1\,k2^2\,k3\,k4\,k5\,k6^2\,k77\,k8\,u^2 + \\
&k11\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k77\,k8\,u^2 + k2^2\,k3\,k33\,k4\,k5\,k6\,k7\,k77\,k8\,u^2 + \\
&k11\,k2\,k3\,k4^2\,k5\,k6\,k7\,k77\,k8\,u^2 + k2^2\,k3\,k4^2\,k5\,k6\,k7\,k77\,k8\,u^2 - \\
&k1\,k2\,k3\,k33\,k4\,k55\,k6\,k7\,k77\,k8\,u^2 - k1\,k2\,k3\,k4^2\,k55\,k6\,k7\,k77\,k8\,u^2 - \\
&k1\,k2\,k3\,k33\,k4\,k6^2\,k7\,k77\,k8\,u^2 - k1\,k2\,k3\,k4^2\,k6^2\,k7\,k77\,k8\,u^2 - k1\,k2^2\,k3\,k33\,k5\,k6^2\,k8^2\,u^2 - \\
&k1\,k2^2\,k3\,k4\,k5\,k6^2\,k8^2\,u^2 + k11\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k8^2\,u^2 + k2^2\,k3\,k33\,k4\,k5\,k6\,k7\,k8^2\,u^2 + \\
&k11\,k2\,k3\,k4^2\,k5\,k6\,k7\,k8^2\,u^2 + k2^2\,k3\,k4^2\,k5\,k6\,k7\,k8^2\,u^2 - k1\,k2\,k3\,k33\,k4\,k55\,k6\,k7\,k8^2\,u^2 - \\
&k1\,k2\,k3\,k4^2\,k55\,k6\,k7\,k8^2\,u^2 - k1\,k2\,k3\,k33\,k4\,k6^2\,k7\,k8^2\,u^2 - \\
&k1\,k2\,k3\,k4^2\,k6^2\,k7\,k8^2\,u^2 + 2\,k1\,k2^2\,k3\,k33\,k4\,k5\,k6\,k7\,k77\,KT\,u^2 + \\
&2\,k1\,k2^2\,k3\,k4^2\,k5\,k6\,k7\,k77\,KT\,u^2 + k1\,k2^2\,k3\,k33\,k5\,k6^2\,k7\,k77\,KT\,u^2 + \\
&k1\,k2^2\,k3\,k4\,k5\,k6^2\,k7\,k77\,KT\,u^2 + 2\,k1\,k2^2\,k3\,k33\,k4\,k5\,k6\,k7\,k8\,KT\,u^2 + \\
&2\,k1\,k2^2\,k3\,k4^2\,k5\,k6\,k7\,k8\,KT\,u^2 + k1\,k2^2\,k3\,k33\,k5\,k6^2\,k7\,k8\,KT\,u^2 + \\
&k1\,k2^2\,k3\,k4\,k5\,k6^2\,k7\,k8\,KT\,u^2 + k1\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k77\,k8\,KT\,u^2 + \\
&k1\,k2\,k3\,k4^2\,k5\,k6\,k7\,k77\,k8\,KT\,u^2 + k1\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k8^2\,KT\,u^2 + \\
&k1\,k2\,k3\,k4^2\,k5\,k6\,k7\,k8^2\,KT\,u^2 - k1\,k2^2\,k3\,k33\,k5\,k6^2\,k7\,k77\,PT\,u^2 - \\
&k1\,k2^2\,k3\,k4\,k5\,k6^2\,k7\,k77\,PT\,u^2 - k1\,k2^2\,k3\,k33\,k5\,k6^2\,k7\,k8\,PT\,u^2 - \\
&k1\,k2^2\,k3\,k4\,k5\,k6^2\,k7\,k8\,PT\,u^2 - k1\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k77\,k8\,PT\,u^2 - \\
&k1\,k2\,k3\,k4^2\,k5\,k6\,k7\,k77\,k8\,PT\,u^2 - 2\,k1\,k2\,k3\,k33\,k5\,k6^2\,k7\,k77\,k8\,PT\,u^2 - \\
&2\,k1\,k2\,k3\,k4\,k5\,k6^2\,k7\,k77\,k8\,PT\,u^2 - k1\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k8^2\,PT\,u^2 - \\
&k1\,k2\,k3\,k4^2\,k5\,k6\,k7\,k8^2\,PT\,u^2 - 2\,k1\,k2\,k3\,k33\,k5\,k6^2\,k7\,k8^2\,PT\,u^2 - \\
&2\,k1\,k2\,k3\,k4\,k5\,k6^2\,k7\,k8^2\,PT\,u^2 + k1\,k2^2\,k3\,k33\,k4\,k55\,k6\,k7\,k77\,u^3 + \\
&k1\,k2^2\,k3\,k4^2\,k55\,k6\,k7\,k77\,u^3 + k1\,k2^2\,k3\,k33\,k4\,k6^2\,k7\,k77\,u^3 + \\
&k1\,k2^2\,k3\,k4^2\,k6^2\,k7\,k77\,u^3 + k1\,k2^2\,k3\,k33\,k4\,k5\,k6\,k77^2\,u^3 + k1\,k2^2\,k3\,k4^2\,k5\,k6\,k77^2\,u^3 - \\
&k1\,k2^2\,k3^2\,k4\,k55\,k6\,k77^2\,u^3 - k1\,k2^2\,k3^2\,k4\,k6^2\,k77^2\,u^3 + k1\,k2^2\,k3\,k33\,k4\,k55\,k6\,k7\,k8\,u^3 + \\
&k1\,k2^2\,k3\,k4^2\,k55\,k6\,k7\,k8\,u^3 + k1\,k2^2\,k3\,k33\,k4\,k6^2\,k7\,k8\,u^3 + k1\,k2^2\,k3\,k4^2\,k6^2\,k7\,k8\,u^3 + \\
&2\,k1\,k2^2\,k3\,k33\,k4\,k5\,k6\,k77\,k8\,u^3 + 2\,k1\,k2^2\,k3\,k4^2\,k5\,k6\,k77\,k8\,u^3 - \\
&2\,k1\,k2^2\,k3^2\,k4\,k55\,k6\,k77\,k8\,u^3 - 2\,k1\,k2^2\,k3^2\,k4\,k6^2\,k77\,k8\,u^3 + \\
&k1\,k2^2\,k3\,k33\,k4\,k5\,k6\,k8^2\,u^3 + k1\,k2^2\,k3\,k4^2\,k5\,k6\,k8^2\,u^3 - k1\,k2^2\,k3^2\,k4\,k55\,k6\,k8^2\,u^3 - \\
&k1\,k2^2\,k3^2\,k4\,k6^2\,k8^2\,u^3 + k1\,k2^2\,k3^2\,k4^2\,k5\,k77^2\,KT\,u^3 + k1\,k2^2\,k3^2\,k4\,k5\,k6\,k77^2\,KT\,u^3 + \\
&2\,k1\,k2^2\,k3^2\,k4^2\,k5\,k77\,k8\,KT\,u^3 + 2\,k1\,k2^2\,k3^2\,k4\,k5\,k6\,k77\,k8\,KT\,u^3 + \\
&k1\,k2^2\,k3^2\,k4^2\,k5\,k8^2\,KT\,u^3 + k1\,k2^2\,k3^2\,k4\,k5\,k6\,k8^2\,KT\,u^3 - k1\,k2^2\,k3^2\,k4\,k5\,k6\,k77^2\,PT\,u^3 - \\
&k1\,k2^2\,k3^2\,k5\,k6^2\,k77^2\,PT\,u^3 - 2\,k1\,k2^2\,k3^2\,k4\,k5\,k6\,k77\,k8\,PT\,u^3 - \\
&2\,k1\,k2^2\,k3^2\,k5\,k6^2\,k77\,k8\,PT\,u^3 - k1\,k2^2\,k3^2\,k4\,k5\,k6\,k8^2\,PT\,u^3 - k1\,k2^2\,k3^2\,k5\,k6^2\,k8^2\,PT\,u^3 + \\
&k1\,k2^2\,k3^2\,k4^2\,k55\,k77^2\,u^4 + k1\,k2^2\,k3^2\,k4^2\,k6\,k77^2\,u^4 + 2\,k1\,k2^2\,k3^2\,k4^2\,k55\,k77\,k8\,u^4 + \\
&2\,k1\,k2^2\,k3^2\,k4^2\,k6\,k77\,k8\,u^4 + k1\,k2^2\,k3^2\,k4^2\,k55\,k8^2\,u^4 + k1\,k2^2\,k3^2\,k4^2\,k6\,k8^2\,u^4))) \Big/ \\
&(k1\,k5\,u\,(k2\,k33\,k6\,k7\,KT + k2\,k4\,k6\,k7\,KT - k33\,k6\,k7\,k8\,PT - k4\,k6\,k7\,k8\,PT + \\
&\quad k2\,k3\,k4\,k77\,KT\,u + k2\,k3\,k4\,k8\,KT\,u - k2\,k3\,k6\,k77\,PT\,u - k2\,k3\,k6\,k8\,PT\,u) \\
&\quad (-k33\,k6\,k7\,k8 - k4\,k6\,k7\,k8 + k2\,k33\,k6\,k7\,u + k2\,k4\,k6\,k7\,u - k2\,k3\,k6\,k77\,u - \\
&\quad k2\,k3\,k6\,k8\,u + k2\,k3\,k4\,k77\,u^2 + k2\,k3\,k4\,k8\,u^2)))
\end{aligned}
$$

In[120]:=

```
Print["x1 flux-balance residual (zero): ", x1RatioResidual];
Print["x2 flux-balance residual (zero): ", x2RatioResidual];
Print["A(u) residual after eliminating the free-enzyme scale (zero): ",
 parametrisationResidual];
Print["ATOT(u), whose level sets enumerate the steady states: "];
Print[TraditionalForm[ATOTofU]];
Print["Positive steady states use u>0
     with A(u)>0 and positive reconstructed concentrations."];
```

x1 flux-balance residual (zero): 0

x2 flux-balance residual (zero): 0

A(u) residual after eliminating the free-enzyme scale (zero): 0

ATOT(u), whose level sets enumerate the steady states:

$$
\begin{aligned}
-\Big(\big((KT - PT\,u\big)\big(&k1\,k2^2\,k3^2\,k4^2\,k55\,k77^2\,u^4 + k1\,k2^2\,k3^2\,k4^2\,k6\,k77^2\,u^4 + \\
&k1\,k2^2\,k3^2\,k4^2\,k55\,k8^2\,u^4 + k1\,k2^2\,k3^2\,k4^2\,k6\,k8^2\,u^4 + 2\,k1\,k2^2\,k3^2\,k4^2\,k55\,k77\,k8\,u^4 + \\
&2\,k1\,k2^2\,k3^2\,k4^2\,k6\,k77\,k8\,u^4 - k1\,k2^2\,k3^2\,k4\,k6^2\,k77^2\,u^3 + k1\,k2^2\,k3\,k4^2\,k5\,k6\,k77^2\,u^3 + \\
&k1\,k2^2\,k3\,k33\,k4\,k5\,k6\,k77^2\,u^3 - k1\,k2^2\,k3^2\,k4\,k55\,k6\,k77^2\,u^3 - k1\,k2^2\,k3^2\,k4\,k6^2\,k8^2\,u^3 + \\
&k1\,k2^2\,k3\,k4^2\,k5\,k6\,k8^2\,u^3 + k1\,k2^2\,k3\,k33\,k4\,k5\,k6\,k8^2\,u^3 - k1\,k2^2\,k3^2\,k4\,k55\,k6\,k8^2\,u^3 + \\
&k1\,k2^2\,k3\,k4^2\,k6^2\,k7\,k77\,u^3 + k1\,k2^2\,k3\,k33\,k4\,k6^2\,k7\,k77\,u^3 + k1\,k2^2\,k3\,k4^2\,k55\,k6\,k7\,k77\,u^3 + \\
&k1\,k2^2\,k3\,k33\,k4\,k55\,k6\,k7\,k77\,u^3 + k1\,k2^2\,k3\,k4^2\,k6^2\,k7\,k8\,u^3 + k1\,k2^2\,k3\,k33\,k4\,k6^2\,k7\,k8\,u^3 + \\
&k1\,k2^2\,k3\,k4^2\,k55\,k6\,k7\,k8\,u^3 + k1\,k2^2\,k3\,k33\,k4\,k55\,k6\,k7\,k8\,u^3 - \\
&2\,k1\,k2^2\,k3^2\,k4\,k6^2\,k77\,k8\,u^3 + 2\,k1\,k2^2\,k3\,k4^2\,k5\,k6\,k77\,k8\,u^3 + \\
&2\,k1\,k2^2\,k3\,k33\,k4\,k5\,k6\,k77\,k8\,u^3 - 2\,k1\,k2^2\,k3^2\,k4\,k55\,k6\,k77\,k8\,u^3 + \\
&k1\,k2^2\,k3^2\,k4^2\,k5\,k77^2\,KT\,u^3 + k1\,k2^2\,k3^2\,k4\,k5\,k6\,k77^2\,KT\,u^3 + k1\,k2^2\,k3^2\,k4^2\,k5\,k8^2\,KT\,u^3 + \\
&k1\,k2^2\,k3^2\,k4\,k5\,k6\,k8^2\,KT\,u^3 + 2\,k1\,k2^2\,k3^2\,k4^2\,k5\,k77\,k8\,KT\,u^3 + \\
&2\,k1\,k2^2\,k3^2\,k4\,k5\,k6\,k77\,k8\,KT\,u^3 - k1\,k2^2\,k3^2\,k5\,k6^2\,k77^2\,PT\,u^3 - \\
&k1\,k2^2\,k3^2\,k4\,k5\,k6\,k77^2\,PT\,u^3 - k1\,k2^2\,k3^2\,k5\,k6^2\,k8^2\,PT\,u^3 - k1\,k2^2\,k3^2\,k4\,k5\,k6\,k8^2\,PT\,u^3 - \\
&2\,k1\,k2^2\,k3^2\,k5\,k6^2\,k77\,k8\,PT\,u^3 - 2\,k1\,k2^2\,k3^2\,k4\,k5\,k6\,k77\,k8\,PT\,u^3 - \\
&k1\,k2^2\,k3\,k33\,k5\,k6^2\,k77^2\,u^2 - k1\,k2^2\,k3\,k4\,k5\,k6^2\,k77^2\,u^2 - k1\,k2^2\,k3\,k33\,k5\,k6^2\,k8^2\,u^2 - \\
&k1\,k2^2\,k3\,k4\,k5\,k6^2\,k8^2\,u^2 - k1\,k2\,k3\,k4^2\,k6^2\,k7\,k8^2\,u^2 - k1\,k2\,k3\,k33\,k4\,k6^2\,k7\,k8^2\,u^2 + \\
&k2^2\,k3\,k4^2\,k5\,k6\,k7\,k8^2\,u^2 + k11\,k2\,k3\,k4^2\,k5\,k6\,k7\,k8^2\,u^2 + k2^2\,k3\,k33\,k4\,k5\,k6\,k7\,k8^2\,u^2 + \\
&k11\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k8^2\,u^2 - k1\,k2\,k3\,k4^2\,k55\,k6\,k7\,k8^2\,u^2 - \\
&k1\,k2\,k3\,k33\,k4\,k55\,k6\,k7\,k8^2\,u^2 + k1\,k2^2\,k33^2\,k5\,k6^2\,k7\,k77\,u^2 + \\
&k1\,k2^2\,k4^2\,k5\,k6^2\,k7\,k77\,u^2 + 2\,k1\,k2^2\,k33\,k4\,k5\,k6^2\,k7\,k77\,u^2 + \\
&k1\,k2^2\,k33^2\,k5\,k6^2\,k7\,k8\,u^2 + k1\,k2^2\,k4^2\,k5\,k6^2\,k7\,k8\,u^2 + 2\,k1\,k2^2\,k33\,k4\,k5\,k6^2\,k7\,k8\,u^2 - \\
&2\,k1\,k2^2\,k3\,k33\,k5\,k6^2\,k77\,k8\,u^2 - 2\,k1\,k2^2\,k3\,k4\,k5\,k6^2\,k77\,k8\,u^2 - \\
&k1\,k2\,k3\,k4^2\,k6^2\,k7\,k77\,k8\,u^2 - k1\,k2\,k3\,k33\,k4\,k6^2\,k7\,k77\,k8\,u^2 + \\
&k2^2\,k3\,k4^2\,k5\,k6\,k7\,k77\,k8\,u^2 + k11\,k2\,k3\,k4^2\,k5\,k6\,k7\,k77\,k8\,u^2 + \\
&k2^2\,k3\,k33\,k4\,k5\,k6\,k7\,k77\,k8\,u^2 + k11\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k77\,k8\,u^2 - \\
&k1\,k2\,k3\,k4^2\,k55\,k6\,k7\,k77\,k8\,u^2 - k1\,k2\,k3\,k33\,k4\,k55\,k6\,k7\,k77\,k8\,u^2 + \\
&k1\,k2\,k3\,k4^2\,k5\,k6\,k7\,k8^2\,KT\,u^2 + k1\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k8^2\,KT\,u^2 + \\
&k1\,k2^2\,k3\,k33\,k5\,k6^2\,k7\,k77\,KT\,u^2 + k1\,k2^2\,k3\,k4\,k5\,k6^2\,k7\,k77\,KT\,u^2 + \\
&2\,k1\,k2^2\,k3\,k4^2\,k5\,k6\,k7\,k77\,KT\,u^2 + 2\,k1\,k2^2\,k3\,k33\,k4\,k5\,k6\,k7\,k77\,KT\,u^2 + \\
&k1\,k2^2\,k3\,k33\,k5\,k6^2\,k7\,k8\,KT\,u^2 + k1\,k2^2\,k3\,k4\,k5\,k6^2\,k7\,k8\,KT\,u^2 +
\end{aligned}
$$

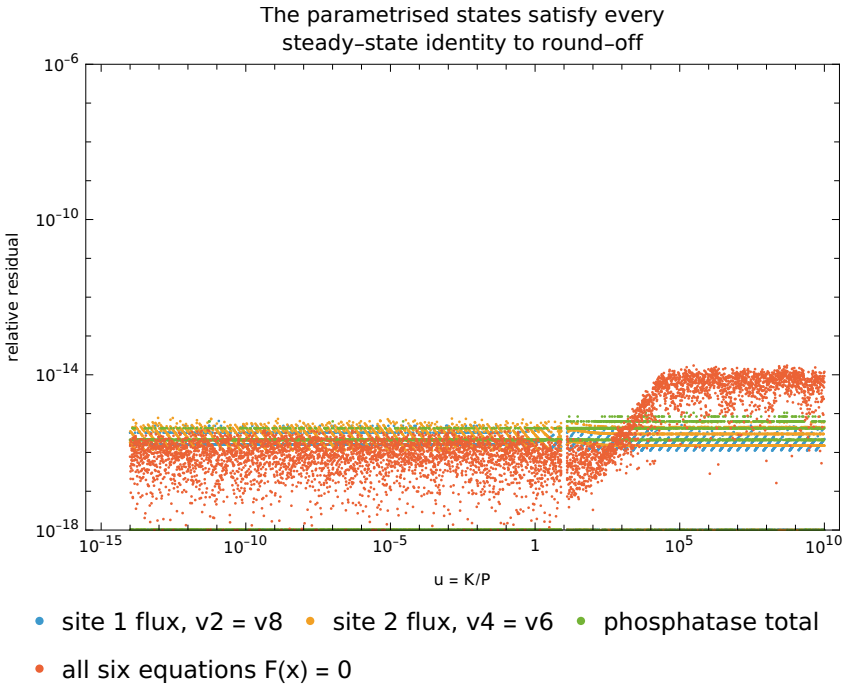
$$
\begin{aligned}
&2\,k1\,k2^2\,k3\,k4^2\,k5\,k6\,k7\,k8\,KT\,u^2 + 2\,k1\,k2^2\,k3\,k33\,k4\,k5\,k6\,k7\,k8\,KT\,u^2 + \\
&k1\,k2\,k3\,k4^2\,k5\,k6\,k7\,k77\,k8\,KT\,u^2 + k1\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k77\,k8\,KT\,u^2 - \\
&2\,k1\,k2\,k3\,k33\,k5\,k6^2\,k7\,k8^2\,PT\,u^2 - 2\,k1\,k2\,k3\,k4\,k5\,k6^2\,k7\,k8^2\,PT\,u^2 - \\
&k1\,k2\,k3\,k4^2\,k5\,k6\,k7\,k8^2\,PT\,u^2 - k1\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k8^2\,PT\,u^2 - \\
&k1\,k2^2\,k3\,k33\,k5\,k6^2\,k7\,k77\,PT\,u^2 - k1\,k2^2\,k3\,k4\,k5\,k6^2\,k7\,k77\,PT\,u^2 - \\
&k1\,k2^2\,k3\,k33\,k5\,k6^2\,k7\,k8\,PT\,u^2 - k1\,k2^2\,k3\,k4\,k5\,k6^2\,k7\,k8\,PT\,u^2 - \\
&2\,k1\,k2\,k3\,k33\,k5\,k6^2\,k7\,k77\,k8\,PT\,u^2 - 2\,k1\,k2\,k3\,k4\,k5\,k6^2\,k7\,k77\,k8\,PT\,u^2 - \\
&k1\,k2\,k3\,k4^2\,k5\,k6\,k7\,k77\,k8\,PT\,u^2 - k1\,k2\,k3\,k33\,k4\,k5\,k6\,k7\,k77\,k8\,PT\,u^2 - \\
&k1\,k2\,k33^2\,k5\,k6^2\,k7\,k8^2\,u - k1\,k2\,k4^2\,k5\,k6^2\,k7\,k8^2\,u - k2^2\,k3\,k33\,k5\,k6^2\,k7\,k8^2\,u - \\
&k11\,k2\,k3\,k33\,k5\,k6^2\,k7\,k8^2\,u - k2^2\,k3\,k4\,k5\,k6^2\,k7\,k8^2\,u - k11\,k2\,k3\,k4\,k5\,k6^2\,k7\,k8^2\,u - \\
&2\,k1\,k2\,k33\,k4\,k5\,k6^2\,k7\,k8^2\,u + k2^2\,k33^2\,k5\,k6^2\,k7^2\,k8\,u + k11\,k2\,k33^2\,k5\,k6^2\,k7^2\,k8\,u + \\
&k2^2\,k4^2\,k5\,k6^2\,k7^2\,k8\,u + k11\,k2\,k4^2\,k5\,k6^2\,k7^2\,k8\,u + 2\,k2^2\,k33\,k4\,k5\,k6^2\,k7^2\,k8\,u + \\
&2\,k11\,k2\,k33\,k4\,k5\,k6^2\,k7^2\,k8\,u - k1\,k2\,k33^2\,k5\,k6^2\,k7\,k77\,k8\,u - k1\,k2\,k4^2\,k5\,k6^2\,k7\,k77\,k8\,u - \\
&k2^2\,k3\,k33\,k5\,k6^2\,k7\,k77\,k8\,u - k11\,k2\,k3\,k33\,k5\,k6^2\,k7\,k77\,k8\,u - \\
&k2^2\,k3\,k4\,k5\,k6^2\,k7\,k77\,k8\,u - k11\,k2\,k3\,k4\,k5\,k6^2\,k7\,k77\,k8\,u - \\
&2\,k1\,k2\,k33\,k4\,k5\,k6^2\,k7\,k77\,k8\,u + k1\,k2^2\,k33^2\,k5\,k6^2\,k7^2\,KT\,u + \\
&k1\,k2^2\,k4^2\,k5\,k6^2\,k7^2\,KT\,u + 2\,k1\,k2^2\,k33\,k4\,k5\,k6^2\,k7^2\,KT\,u + k1\,k2\,k33^2\,k5\,k6^2\,k7^2\,k8\,KT\,u + \\
&k1\,k2\,k4^2\,k5\,k6^2\,k7^2\,k8\,KT\,u + 2\,k1\,k2\,k33\,k4\,k5\,k6^2\,k7^2\,k8\,KT\,u - k1\,k33^2\,k5\,k6^2\,k7^2\,k8^2\,PT\,u - \\
&k1\,k4^2\,k5\,k6^2\,k7^2\,k8^2\,PT\,u - 2\,k1\,k33\,k4\,k5\,k6^2\,k7^2\,k8^2\,PT\,u - k1\,k2\,k33^2\,k5\,k6^2\,k7^2\,k8\,PT\,u - \\
&k1\,k2\,k4^2\,k5\,k6^2\,k7^2\,k8\,PT\,u - 2\,k1\,k2\,k33\,k4\,k5\,k6^2\,k7^2\,k8\,PT\,u - \\
&k11\,k33^2\,k5\,k6^2\,k7^2\,k8^2 - k2\,k33^2\,k5\,k6^2\,k7^2\,k8^2 - k11\,k4^2\,k5\,k6^2\,k7^2\,k8^2 - \\
&k2\,k4^2\,k5\,k6^2\,k7^2\,k8^2 - 2\,k11\,k33\,k4\,k5\,k6^2\,k7^2\,k8^2 - 2\,k2\,k33\,k4\,k5\,k6^2\,k7^2\,k8^2\big)\big)\big/ \\
&\big(k1\,k5\,u\,(k2\,k33\,k6\,k7\,KT + k2\,k4\,k6\,k7\,KT + k2\,k3\,k4\,k77\,u\,KT + k2\,k3\,k4\,k8\,u\,KT - \\
&\quad k33\,k6\,k7\,k8\,PT - k4\,k6\,k7\,k8\,PT - k2\,k3\,k6\,k77\,PT\,u - k2\,k3\,k6\,k8\,PT\,u\,) \\
&\quad (k2\,k3\,k4\,k77\,u^2 + k2\,k3\,k4\,k8\,u^2 + k2\,k33\,k6\,k7\,u + k2\,k4\,k6\,k7\,u - \\
&\quad k2\,k3\,k6\,k77\,u - k2\,k3\,k6\,k8\,u - k33\,k6\,k7\,k8 - k4\,k6\,k7\,k8)\big)\big)
\end{aligned}
$$

Positive steady states use u>0 with A(u)>0 and positive reconstructed concentrations.

Added figure. Plot: along every branch of the Fig 3 set, the relative residuals of the two site-flux balances (v2 = v8, v4 = v6), of the phosphatase total, and of all six steady-state equations F(x) = 0, evaluated numerically for the reconstructed states. All lie at round-off level, so every grid point of the parametrisation is an exact steady state. Saved as S6_parametrisation_residuals.pdf.

In[126]:=

```
(* Plot for Section 6: numerical residuals of the identities along the parametrised brar
Module[{pr = figSet3, pts, res, fl = 10^-18},
 pts = Flatten[figBranches[pr], 1];
 res = Function[s, With[{xx = s["x"], pp = s["P"]},
      {s["u"],
       Max[fl, Abs[pr["k2"] xx[[3]] - pr["k8"] xx[[6]]]/(pr["k2"] xx[[3]])],
       Max[fl, Abs[pr["k4"] xx[[4]] - pr["k6"] xx[[5]]]/(pr["k4"] xx[[4]])],
       Max[fl, Abs[pp + xx[[5]] + xx[[6]] - pr["PTOT"]]/pr["PTOT"]],
       Max[fl, Max[Abs[figRHS[pr, s["ATOT"], xx]]]/Max[pr["k1"] s["ATOT"] pr["KTOT"], Max[Abs[x
 With[{g = ListLogLogPlot[{res[[All, {1, 2}]], res[[All, {1, 3}]], res[[All, {1, 4}]], res[[All,
   PlotRange -> {All, {10^-18, 10^-6}}, Frame -> True, FrameLabel -> {"u = K/P", "relative
   PlotLegends -> Placed[{"site 1 flux, v2 = v8", "site 2 flux, v4 = v6", "phosphatase t
   PlotLabel -> Column[{"The parametrised states satisfy every", "steady-state identity
  figShow["S6_parametrisation_residuals", {g}, g]]]
```




```
saved:
  /wolframcloud/userfiles/9ff/9ffe219c-88d9-4035-be0c-26993d670000/Files/Harika/Derivation_Notebooks/Final/plots/S6_parametrisation
  _residuals.pdf
```

7. Derive fold, detached-branch, and branch-point conditions

In[127]:=

```
(* Fold, detached-branch, and branch-point conditions *)

(*
  Companion to main-text Eqs. (10)--(11) and the associated Methods and
  Results discussion. The exact branch is parameterised by u=K/P, so folds
  are stationary points of ATOT(u); branch endpoints are the sign-changing
  factors u=rho and u*= -beta/alpha.
*)
```

A(u) has endpoint factors at u = r and u* = $-\beta/\alpha$. The strict inequalities are the analytical detached-branch condition. Under the symmetric iteration constraints, the crossing occurs at u = 1; the factorisation residual is printed as a diagnostic.

In[128]:=
```
(* Keep rho, m1, m3, and m7 as independent symbols so the symmetry
   substitutions below remain explicit. Their kinetic definitions are
   m1=(k11+k2)/k1, m3=(k33+k4)/k3, m7=(k77+k8)/k7. *)
a1 = k2 m7 / (k8 m1);
alpha = (a1 / m3) (rho k4 / k6 - 1);
beta = (1 / m1) (rho k2 / k8 - 1);
```

In[131]:=
```
(* The second possible zero of A(u), reduced to the explicit form in Eq. (11). *)
uStarFromAlphaBeta = Cancel[-beta / alpha]
uStarEq11 = m3 k8 (1 - rho k2 / k8) / (m7 k2 (rho k4 / k6 - 1))
uStarResidual = Together[uStarFromAlphaBeta - uStarEq11]
```

Out[131]=
$$\frac{k6\ m3\ (-k8 + k2\ rho)}{k2\ m7\ (k6 - k4\ rho)}$$

Out[132]=
$$\frac{k8\ m3\left(1 - \frac{k2\ rho}{k8}\right)}{k2\ m7\left(-1 + \frac{k4\ rho}{k6}\right)}$$

Out[133]=
```
0
```

In[134]:=
```
(* Detached branch condition (strict inequalities; equality is a degenerate
   boundary): k8 PT > k2 KT and k4 KT > k6 PT, with u* ≠ rho. *)
detachedBranchCondition = k8 PT > k2 KT && k4 KT > k6 PT
```

Out[134]=
```
k8 PT > k2 KT && k4 KT > k6 PT
```

In[135]:=
```
(* Iterations 3--4: equal enzyme totals and matched second/outer-step
   constants imply m7=m3, k6=k2, k8=k4, and rho=1. *)
symmetricRules = {rho → 1, k6 → k2, k8 → k4, m7 → m3};
uStarAtSymmetry = Cancel[uStarEq11 /. symmetricRules]
alphaAtSymmetry = Cancel[alpha /. symmetricRules]
betaAtSymmetry = Cancel[beta /. symmetricRules]
```

Out[136]=
```
1
```

Out[137]=
$$\frac{-k2 + k4}{k4\ m1}$$

Out[138]=
$$\frac{k2 - k4}{k4\ m1}$$

In[139]:=

```
(* The scalar enzyme-balance equation before dividing by its factors.
   Its numerator displays the two intersecting steady-state branches. *)
balanceEquation = rho -
 u (1 + A / m1 + a1 u A / m3) /
  (1 + (a1 k4 / (k6 m3)) u ^ 2 A + a1 u A / m7);
balanceNumeratorAtSymmetry = Factor[
 Numerator[Together[balanceEquation /. symmetricRules]]]
branchFactorAtSymmetry = Factor[
 (u - 1) (alphaAtSymmetry u A - 1)]
factorisationResidual = Factor[
 balanceNumeratorAtSymmetry - branchFactorAtSymmetry]
```

Out[140]=

$$-((-1+u)\,(k4\ m1 + A\ k2\ u - A\ k4\ u))$$

Out[141]=

$$\frac{(-1+u)\,(-k4\ m1 - A\ k2\ u + A\ k4\ u)}{k4\ m1}$$

Out[142]=

$$\frac{(-1+k4\ m1)\,(-1+u)\,(-k4\ m1 - A\ k2\ u + A\ k4\ u)}{k4\ m1}$$

In[143]:=

```
(* At the crossing, alpha>0 exactly when k4>k2; the branch point has
   u=1 and A=1/alpha. *)
branchPointExistsCondition = k4 > k2;
branchPointFreeSubstrate = 1/alphaAtSymmetry
```

Out[144]=

$$\frac{k4\ m1}{-k2 + k4}$$

In[145]:=

```
Print["u* formula residual (zero): ", uStarResidual];
Print["u* under the symmetric-iteration constraints: ", uStarAtSymmetry];
Print["alpha under those constraints: ", alphaAtSymmetry];
Print["beta + alpha (zero under the constraints): ",
 FullSimplify[alphaAtSymmetry + betaAtSymmetry]];
Print["Crossing-branch factorisation residual (zero): ",
 factorisationResidual];
Print["Detached-branch condition: k8 PT > k2 KT and k4 KT > k6 PT."];
Print["Branch-point condition under symmetry: k4 > k2."];
```

u* formula residual (zero): 0

u* under the symmetric-iteration constraints: 1

alpha under those constraints: $\frac{-k2 + k4}{k4\ m1}$

beta + alpha (zero under the constraints): 0

Crossing-branch factorisation residual (zero): $\frac{(-1 + k4\ m1)\,(-1 + u)\,(-k4\ m1 - A\ k2\ u + A\ k4\ u)}{k4\ m1}$

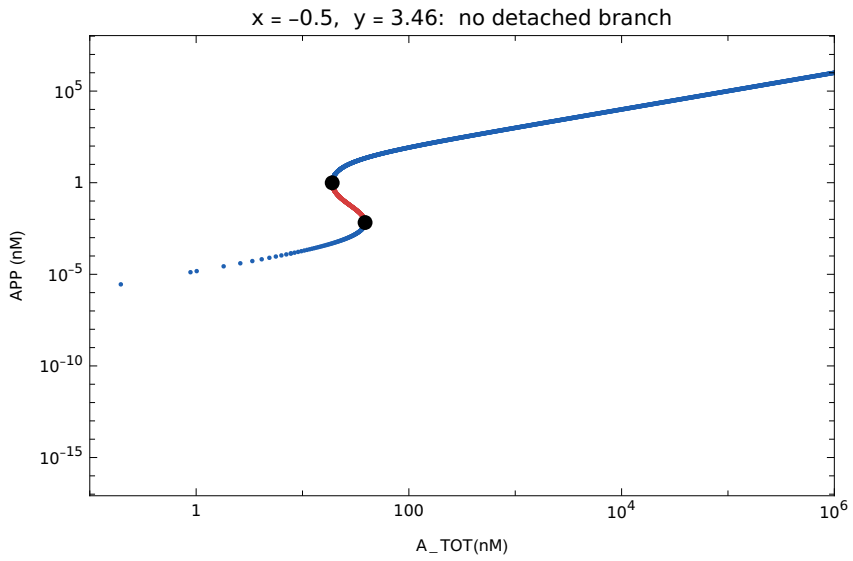
Detached-branch condition: k8 PT > k2 KT and k4 KT > k6 PT.

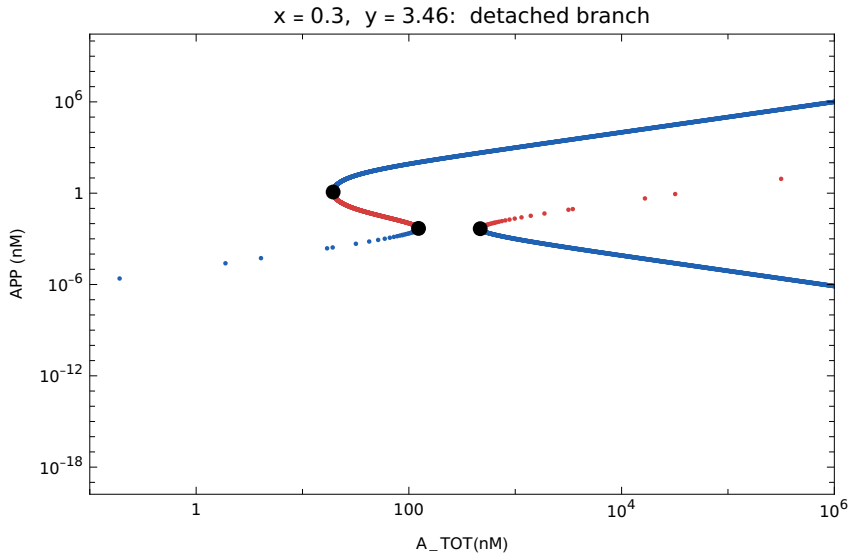
Branch-point condition under symmetry: k4 > k2.

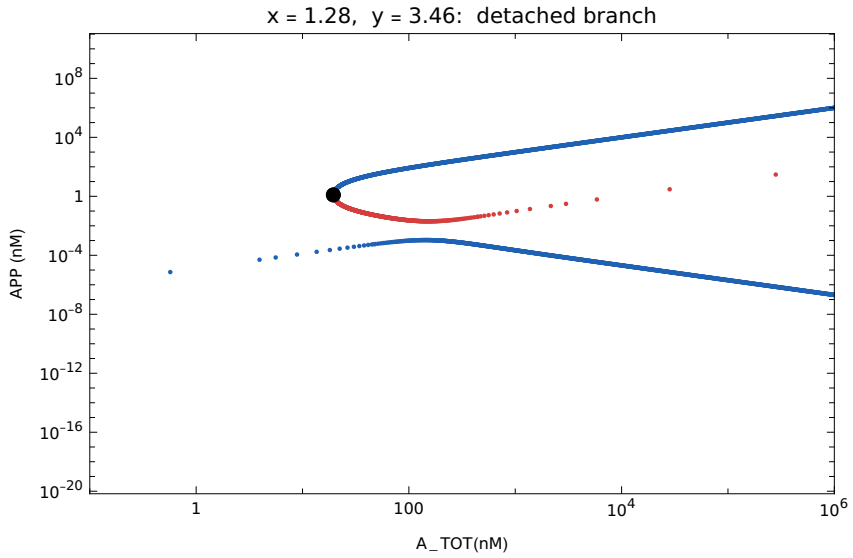
Added figure. Plots: (top) the Fig 3 set with k8 changed so that x = log10(k8 P_TOT/(k2 K_TOT)) = -0.5, 0.3 and its original value 1.28, at fixed y = log10(k4 K_TOT/(k6 P_TOT)) = 3.46; with x < 0 criterion (10) fails and there is no detached branch (Class 1), with x > 0 a detached branch appears (Classes 2 and 5); the last panel shows the three sets in the (x, y) plane. (Bottom) the symmetric Class 6 set with k4 > k2 (branch point at u = 1, A = 1/alpha) and the same set with k4 = k8 = k2/2 (alpha < 0: no branch point). Saved as S7_detached_branch_and_branch_point_conditions.pdf.

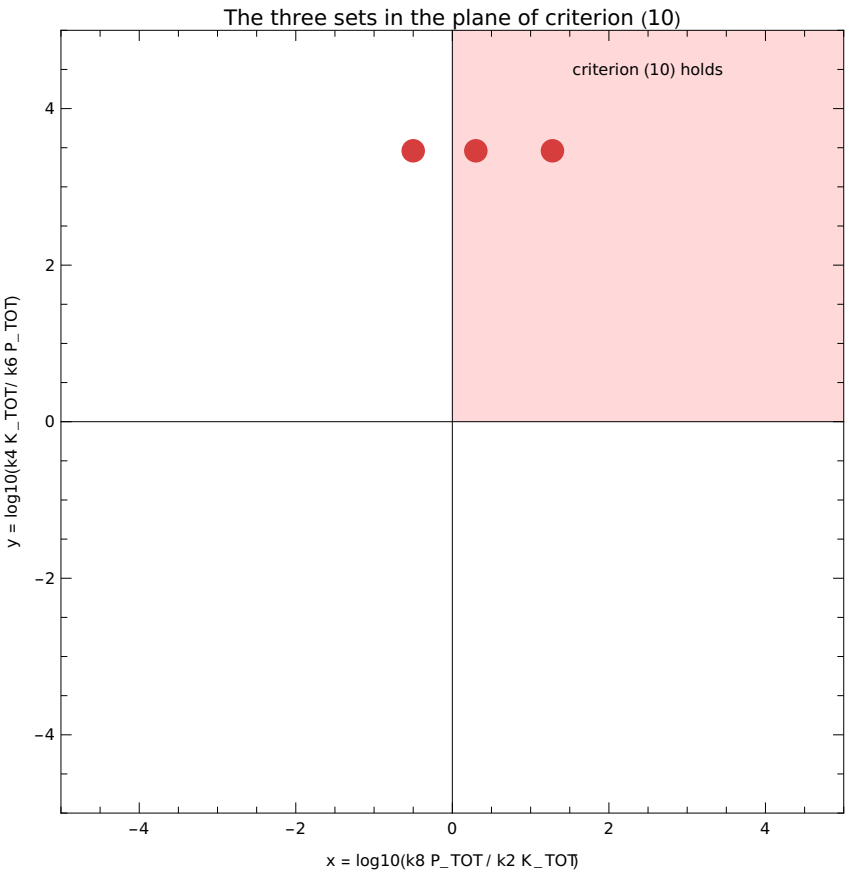
In[152]:=

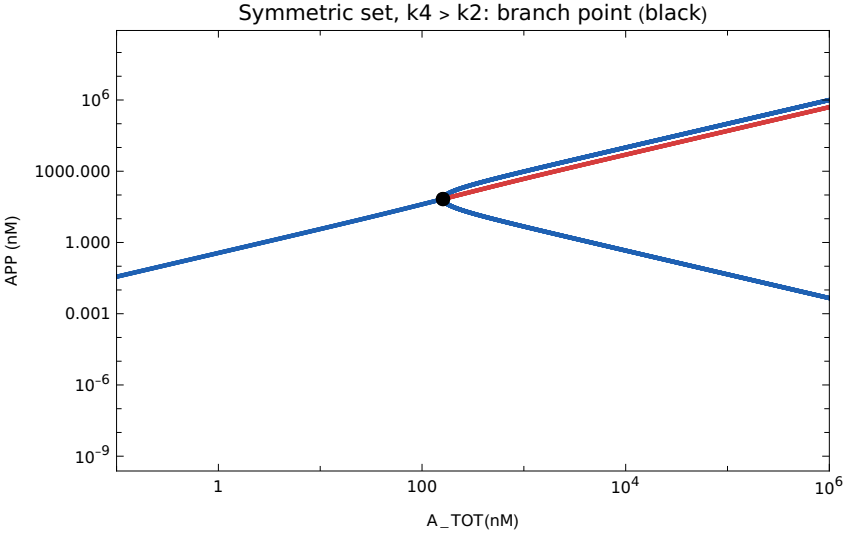
```
(* Plot for Section 7: criterion (10) for the detached branch and the branch-point condit
Module[{pr = figSet3, x0, y0, xt, sets, diags, plane, symNo, d6, d6no},
 {x0, y0} = {Log10[pr["k8"] pr["PTOT"]/(pr["k2"] pr["KTOT"])], Log10[pr["k4"] pr["KTOT"]/(pr["k6"]
 xt = {-0.5, 0.3, x0};
 sets = Function[xv, Module[{q = pr}, q["k8"] = pr["k8"] 10^(xv - x0); q]] /@ xt;
 diags = MapThread[figDiagram[#1, PlotLabel -> ("x = " <> figNum[#2, 3] <> ",  y = " <> f
        If[#2 > 0 && y0 > 0, ":  detached branch", ":  no detached branch"]), ImageSize
 plane = Graphics[{RGBColor[1, 0.85, 0.85], Rectangle[{0, 0}, {5, 5}], Black, Line[{{-5, 0},
    Line[{{0, -5}, {0, 5}}], figRed, PointSize[0.03], Point[{#, y0} & /@ xt],
    Black, Text["criterion (10) holds", {2.5, 4.5}]},
   Frame -> True, PlotRange -> {{-5, 5}, {-5, 5}}, AspectRatio -> 1,
   FrameLabel -> {"x = log10(k8 P_TOT / k2 K_TOT)", "y = log10(k4 K_TOT / k6 P_TOT)"},
   PlotLabel -> "The three sets in the plane of criterion (10)", ImageSize -> 440];
 symNo = Module[{q = figSet6}, q["k4"] = figSet6["k2"]/2; q["k8"] = figSet6["k2"]/2; q];
 d6 = figDiagram[figSet6, PlotLabel -> "Symmetric set, k4 > k2: branch point (black)", Ir
 d6no = figDiagram[symNo, PlotLabel -> "Same set with k4 = k8 = k2/2: no branch point",
 figShow["S7_detached_branch_and_branch_point_conditions",Join[diags, {plane, d6, d6no}]
  Column[{GraphicsRow[Append[diags, plane], ImageSize -> 1300], GraphicsRow[{d6, d6no}, Ima
```

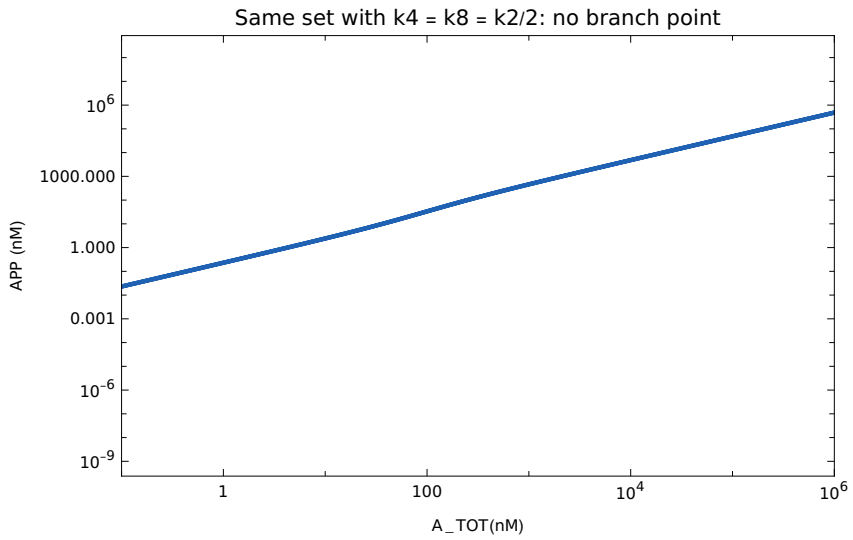

x = –0.5, y = 3.46: no detached branch
APP (nM)
10^{5}
1
10^{-5}
10^{-10}
10^{-15}
1
100
10^{4}
10^{6}
A_TOT(nM)


x = 0.3, y = 3.46: detached branch
APP (nM)
10^{6}
1
10^{-6}
10^{-12}
10^{-18}
1
100
10^{4}
10^{6}
A_TOT(nM)


x = 1.28, y = 3.46: detached branch
APP (nM)
10^{8}
10^{4}
1
10^{-4}
10^{-8}
10^{-12}
10^{-16}
10^{-20}
1
100
10^{4}
10^{6}
A_TOT(nM)


The three sets in the plane of criterion (10)
criterion (10) holds
y = log10(k4 K_TOT/ k6 P_TOT)
x = log10(k8 P_TOT / k2 K_TOT)
4
2
0
−2
−4
−4
−2
0
2
4


Symmetric set, k4 > k2: branch point (black)
APP (nM)
A_TOT(nM)
10^6
1000.000
1.000
0.001
10^-6
10^-9
1
100
10^4
10^6



saved:
/wolframcloud/userfiles/9ff/9ffe219c-88d9-4035-be0c-26993d670000/Files/Harika/Derivation_Notebooks/Final/plots/S7_detached_branch_and_branch_point_conditions.pdf

8. Derive the crossing-branch slope for Classes 7 and 8

In[153]:=

```
(* Crossing-branch slope and the analytical Class 7 / Class 8 criterion *)

(*
  Companion to main-text Eqs. (12)--(14) and Methods, "Slope of the crossing
  branch, Eq. (13)". This derives the logarithmic slope of ATOT(u) at the
  branch point under the Iteration-3/4 constraints.
*)
```

With equal totals and matched second-step rates, the logarithmic slope at the crossing has the sign of M5 – M1. This gives the analytical Class 7/Class 8 criterion and the symmetric pitchfork limit.

In[154]:=

```
(* Constraints: KT=PT, k6=k2, k8=k4, k7=k3, k77=k33, hence M7=M3.
   First-binding constants remain free, so M1 and M5 need not be equal. *)
a1 = k2 M3/(k4 M1);
a1a2 = M5/M1;
a2 = a1a2/a1;
alpha = (k4 - k2)/(k4 M1);
```

In[158]:=

```
(* Along the crossing branch A=1/(alpha u), reconstruct x1...x6 and K. *)
KofU = KT/(1 + 1/(alpha M1 u) + a1/(alpha M3));
x1ofU = a1/alpha;
x2ofU = a1a2 u/alpha;
x3ofU = KofU/(alpha M1 u);
x4ofU = a1 KofU/(alpha M3);
x5ofU = a1a2 KofU/(alpha M5);
x6ofU = a1 KofU/(alpha M3 u);
```

In[165]:=

```
ATOTofU = Together[1/(alpha u) + x1ofU + x2ofU + x3ofU +
  x4ofU + x5ofU + x6ofU]
```

Out[165]=

$$\frac{1}{k1\ k3\ (k2-k4)\ k4\ k5\ u}\left(-k11\ k3\ k4^2\ k5-k2\ k3\ k4^2\ k5-k1\ k2\ k33\ k4\ k5\ u-\right.$$
$$\left.k1\ k2\ k4^2\ k5\ u+k1\ k2^2\ k3\ k5\ KT\ u-k1\ k3\ k4^2\ k5\ KT\ u-k1\ k3\ k4^2\ k55\ u^2-k1\ k3\ k4^2\ k6\ u^2\right)$$

In[166]:=

```
(* Eq. (12): d ln(ATOT) / d ln(u) at u=1. *)
slopeFromParametrisation = FullSimplify[
 (u D[ATOTofU, u]/ ATOTofU) /. u → 1]
slopeEq13 = k4^2 (M5 - M1)/(KT (k4^2 - k2^2) + k4^2 (M1 + M5) + k2 k4 M3)
slopeResidual = Together[slopeFromParametrisation - slopeEq13]
```

Out[166]=

$$\frac{k3\ k4^2\ (-((k11+k2)\ k5)+k1\ (k55+k6))}{k4\ ((k11+k2)\ k3\ k4\ k5+k1\ k2\ (k33+k4)\ k5+k1\ k3\ k4\ (k55+k6))+k1\ k3\ (-k2+k4)\ (k2+k4)\ k5\ KT}$$

Out[167]=

$$\frac{k4^2\left(-\frac{k11+k2}{k1}+\frac{k55+k6}{k5}\right)}{\frac{k2\ k4\ (k33+k4)}{k3}+k4^2\left(\frac{k11+k2}{k1}+\frac{k55+k6}{k5}\right)+\left(-k2^2+k4^2\right)KT}$$

Out[168]=

```
0
```

In[169]:=

```
(* For positive totals/constants and k4>k2, the denominator is positive.
   Thus sign(s)=sign(M5-M1): Class 7 iff M5<M1; Class 8 iff M5>M1.
   When M5=M1, s=0 and the crossing is the symmetric pitchfork limit. *)
positiveDenominator = KT (k4^2 - k2^2) +
 k4^2 (M1 + M5) + k2 k4 M3
```

Out[169]=

$$\frac{k2\ k4\ (k33+k4)}{k3}+k4^2\left(\frac{k11+k2}{k1}+\frac{k55+k6}{k5}\right)+\left(-k2^2+k4^2\right)KT$$

In[170]:=

```
Print["ATOT(u) on the crossing branch: "];
Print[TraditionalForm[ATOTofU]];
Print["Slope formula residual (zero): ", slopeResidual];
Print["Eq. (13) slope: ", slopeEq13];
Print["Class 7 condition: M5 < M1;
    Class 8 condition: M5 > M1; pitchfork limit: M5 = M1."];
```

ATOT(u) on the crossing branch:

$$\frac{1}{k1\,k3\,k4\,k5\,u\,(k2-k4)}\left(k1\,k2^2\,k3\,k5\,KT\,u - k1\,k2\,k33\,k4\,k5\,u - k1\,k2\,k4^2\,k5\,u - k1\,k3\,k4^2\,k5\,KT\,u - k1\,k3\,k4^2\,k55\,u^2 - k1\,k3\,k4^2\,k6\,u^2 - k11\,k3\,k4^2\,k5 - k2\,k3\,k4^2\,k5\right)$$

Slope formula residual (zero): 0

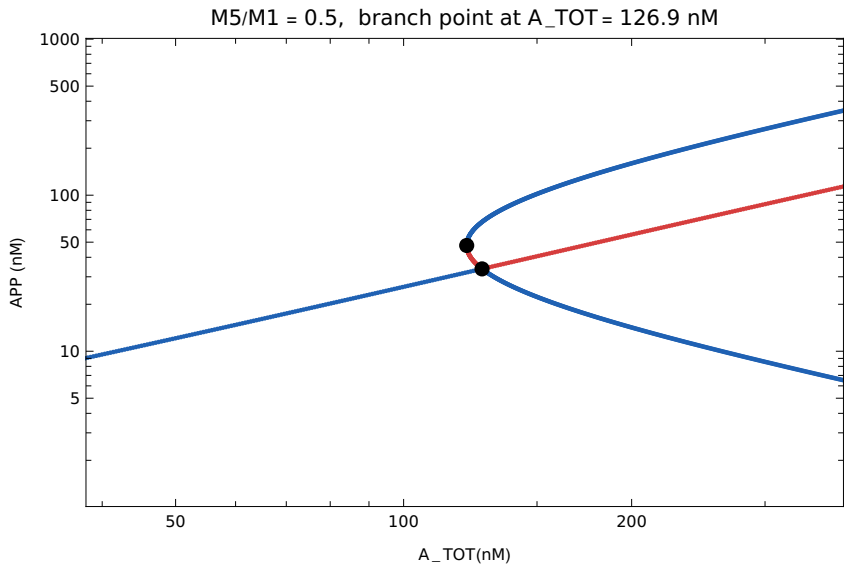
Eq. (13) slope: $$\frac{k4^2\left(-\frac{k11+k2}{k1}+\frac{k55+k6}{k5}\right)}{\frac{k2\,k4\,(k33+k4)}{k3}+k4^2\left(\frac{k11+k2}{k1}+\frac{k55+k6}{k5}\right)+\left(-k2^2+k4^2\right)KT}$$

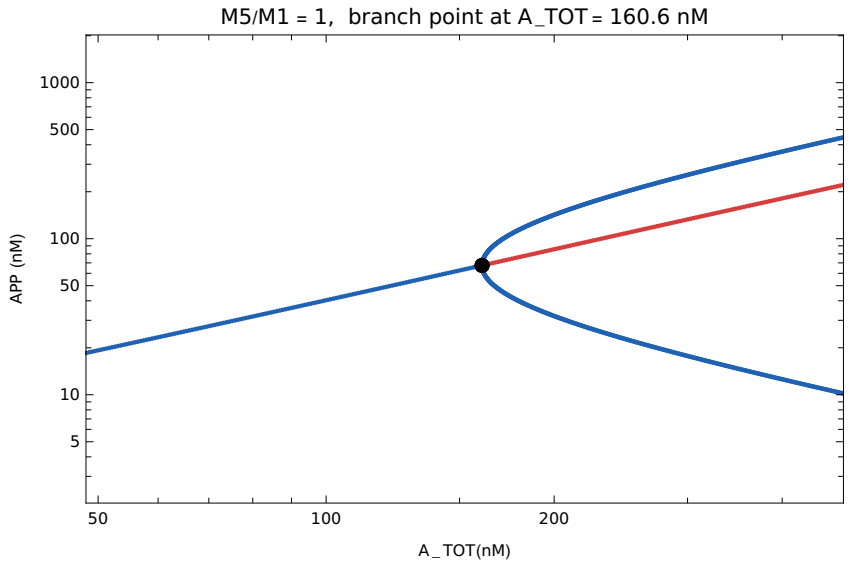
Class 7 condition: M5 < M1; Class 8 condition: M5 > M1; pitchfork limit: M5 = M1.

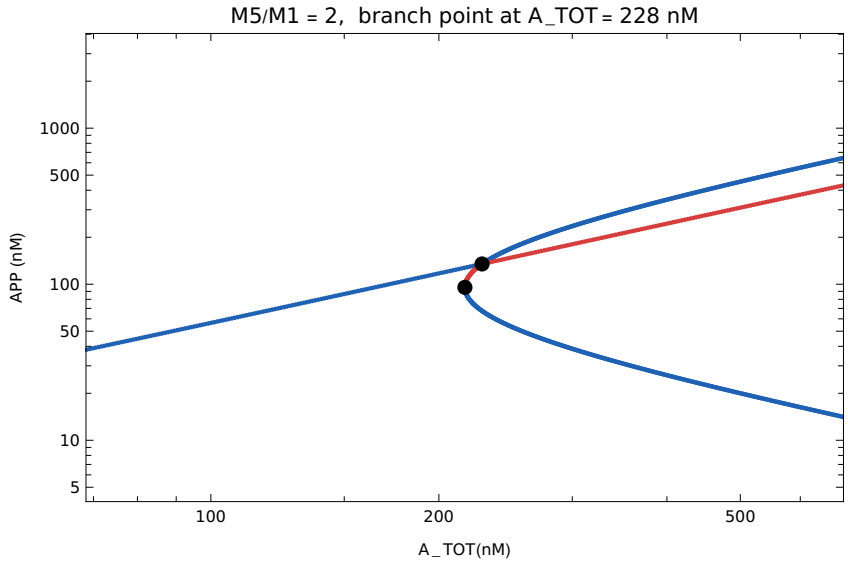
Added figure. Plots: the symmetric Class 6 set with k5 scaled so that M5/M1 = 1/2, 1 and 2 (first-binding asymmetry). (Top) the diagram near the branch point: a pitchfork for M5 = M1 and transcritical crossings of opposite orientation for M5 < M1 (Class 7) and M5 > M1 (Class 8). (Bottom) the slope s of Eq. (13) as a function of log10(M5/M1) for this set (line), with the slope d ln A_TOT / d ln u at u = 1 computed numerically from the parametrisation (points). Saved as S8_crossing_branch_slope.pdf.

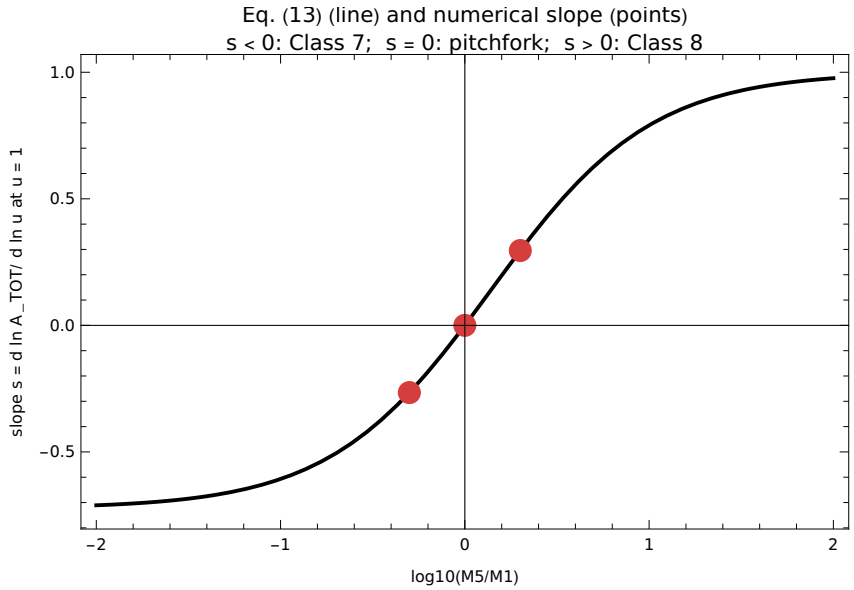
In[175]:=

```
(* Plot for Section 8: orientation of the crossing branch, slope s of Eq. (13) *)
Module[{base = figSet6, f = {2., 1., 0.5}, sets, zooms, c0, sEq13, sNum, g},
 sets = Function[fv, Module[{q = base}, q["k5"] = base["k5"] fv; q]] /@ f;
 zooms = MapThread[figDiagram[#1, PlotRange -> {{0.3, 3} #2["ATOT"], {0.03, 30} #2["x"][[2]]
       PlotLabel -> ("M5/M1 = " <> figNum[figConst[#1]["M5"]/figConst[#1]["M1"], 3] <>
          ",  branch point at A_TOT = " <> figNum[#2["ATOT"]] <> " nM"), ImageSize -> 440
 c0 = figConst[base];
 sEq13[lr_] := With[{m1 = c0["M1"], m5 = 10^lr c0["M1"], m3 = c0["M3"], k2 = base["k2"], k4 =
   k4^2 (m5 - m1)/(kt (k4^2 - k2^2) + k4^2 (m1 + m5) + k2 k4 m3)];
 sNum[q_Association] := With[{c = figConst[q], hh = 10^-5},
   (Log[figState[q, c, 1 + hh, 1/(c["alpha"] (1 + hh))]["ATOT"]] -
     Log[figState[q, c, 1 - hh, 1/(c["alpha"] (1 - hh))]["ATOT"]])/(2 hh)];
 g = Show[Plot[sEq13[lr], {lr, -2, 2}, PlotStyle -> Black],
   ListPlot[{Log10[figConst[#]["M5"]/figConst[#]["M1"]], sNum[#]} & /@ sets, PlotStyle -> Dir
   Frame -> True, GridLines -> {{0}, {0}}, FrameLabel -> {"log10(M5/M1)", "slope s = d ln A
   PlotLabel -> Column[{"Eq. (13) (line) and numerical slope (points)", "s < 0: Class 7;  s
   ImageSize -> 440];
 figShow["S8_crossing_branch_slope",Append[zooms, g], Column[{GraphicsRow[zooms, ImageSiz
```


M5/M1 = 0.5, branch point at A_TOT = 126.9 nM
APP (nM)
1000
500
100
50
10
5
50
100
200
A_TOT(nM)


M5/M1 = 1, branch point at A_TOT = 160.6 nM
APP (nM)
1000
500
100
50
10
5
50
100
200
A_TOT(nM)


M5/M1 = 2, branch point at A_TOT = 228 nM
APP (nM)
1000
500
100
50
10
5
100
200
500
A_TOT(nM)



saved:
/wolframcloud/userfiles/9ff/9ffe219c-88d9-4035-be0c-26993d670000/Files/Harika/Derivation_Notebooks/Final/plots/S8_crossing_branch_slope.pdf

9. Define fold and branch-point normal-form coefficients

In[176]:=

```
(* Fold and branch-point normal-form coefficient definitions *)

(*
  Companion to main-text Methods, "Analytic derivatives and normal-form
  coefficients", and Supplementary Note 4. These are the contractions used
  by the MATLAB normal_forms.m implementation. At a singular steady state,
  supply the left/right nullvectors p and q.
*)
```

These are symbolic coefficient definitions. Numerical values require a singular state and the corresponding left and right nullvectors from the bifurcation calculation.

In[177]:=

```
state = {x1, x2, x3, x4, x5, x6};
augmentedState = Join[state, {ATOT}];
freeA = ATOT - Total[state];
freeK = KTOT - x3 - x4;
freeP = PTOT - x5 - x6;
```

In[182]:=
```
v1 = k1 freeA freeK - k11 x3; v2 = k2 x3;
v3 = k3 x1 freeK - k33 x4; v4 = k4 x4;
v5 = k5 x2 freeP - k55 x5; v6 = k6 x5;
v7 = k7 x1 freeP - k77 x6; v8 = k8 x6;
F = {v2 - v3 + v6 - v7, v4 - v5, v1 - v2,
 v3 - v4, v5 - v6, v7 - v8}
```

Out[186]=
$$\{k2\,x3 - k3\,x1\,(KTOT - x3 - x4) + k33\,x4 + k6\,x5 - k7\,x1\,(PTOT - x5 - x6) + k77\,x6,\\ k4\,x4 + k55\,x5 - k5\,x2\,(PTOT - x5 - x6),\\ -k11\,x3 - k2\,x3 + k1\,(KTOT - x3 - x4)\,(ATOT - x1 - x2 - x3 - x4 - x5 - x6),\\ k3\,x1\,(KTOT - x3 - x4) - k33\,x4 - k4\,x4,\\ -k55\,x5 - k6\,x5 + k5\,x2\,(PTOT - x5 - x6),\ k7\,x1\,(PTOT - x5 - x6) - k77\,x6 - k8\,x6\}$$

In[187]:=
```
J = Table[D[F[[i]], state[[j]]], {i, 6}, {j, 6}];
FATOT = D[F, ATOT]
```

Out[188]=
$$\{0,\ 0,\ k1\,(KTOT - x3 - x4),\ 0,\ 0,\ 0\}$$

In[189]:=
```
(* B(v,w)=D^2F[(v),(w)] in augmented (state, ATOT) coordinates. *)
B[vec1_List, vec2_List] :=
 Table[Sum[D[F[[i]], augmentedState[[j]], augmentedState[[k]]] * vec1[[j]] * vec2[[k]],
   {j, 7}, {k, 7}], {i, 6}]
```

In[190]:=
```
(* Fold: J q=0 and p^T J=0. With ||q||=1 and p^T q=1,
   aLP is the quadratic normal-form coefficient and betaLP the
   parameter-transversality coefficient. *)
qNull = Array[q, 6]; pNull = Array[p, 6];
Qfold = Join[qNull, {0}];
aLP = (pNull . B[Qfold, Qfold])/2;
betaLP = pNull . FATOT;
kappaLP = -aLP/betaLP
```

Out[194]=
$$\frac{1}{2\,k1\,(KTOT - x3 - x4)\,p[3]}\Big(-p[4]\,(-2\,k3\,q[1]\times q[3] - 2\,k3\,q[1]\times q[4]) - p[6]\,(-2\,k7\,q[1]\times q[5] - 2\,k7\,q[1]\times q[6]) -\\ p[1]\,(2\,k3\,q[1]\times q[3] + 2\,k3\,q[1]\times q[4] + 2\,k7\,q[1]\times q[5] + 2\,k7\,q[1]\times q[6]) -\\ p[5]\,(-2\,k5\,q[2]\times q[5] - 2\,k5\,q[2]\times q[6]) - p[2]\,(2\,k5\,q[2]\times q[5] + 2\,k5\,q[2]\times q[6]) -\\ p[3]\,\big(2\,k1\,q[1]\times q[3] + 2\,k1\,q[2]\times q[3] + 2\,k1\,q[3]^2 + 2\,k1\,q[1]\times q[4] + 2\,k1\,q[2]\times q[4] +\\ 4\,k1\,q[3]\times q[4] + 2\,k1\,q[4]^2 + 2\,k1\,q[3]\times q[5] + 2\,k1\,q[4]\times q[5] + 2\,k1\,q[3]\times q[6] + 2\,k1\,q[4]\times q[6]\big)\Big)$$

In[195]:=

```
(* Branch point: solve J v0 + p lambda = -F_ATOT and q^T v0=0.
   The local branches satisfy a xi^2 + 2 b xi eta + c eta^2=0. *)
v0 = Array[v, 6];
Qbp = Join[qNull, {0}];
Vbp = Join[v0, {1}];
aBP = pNull . B[Qbp, Qbp];
bBP = pNull . B[Qbp, Vbp];
cBP = pNull . B[Vbp, Vbp];
```

In[201]:=

```
branchDiscriminant = bBP^2 - aBP cBP
branchingEquation = aBP xi^2 + 2 bBP xi eta + cBP eta^2
```

Out[201]=

$$
\begin{aligned}
&(p[4]\,(-k3\,q[3]\times v[1]-k3\,q[4]\times v[1]-k3\,q[1]\times v[3]-k3\,q[1]\times v[4])+\\
&\quad p[6]\,(-k7\,q[5]\times v[1]-k7\,q[6]\times v[1]-k7\,q[1]\times v[5]-k7\,q[1]\times v[6])+\\
&\quad p[1]\,(k3\,q[3]\times v[1]+k3\,q[4]\times v[1]+k7\,q[5]\times v[1]+k7\,q[6]\times v[1]+k3\,q[1]\times v[3]+k3\,q[1]\times v[4]+\\
&\qquad k7\,q[1]\times v[5]+k7\,q[1]\times v[6])+p[5]\,(-k5\,q[5]\times v[2]-k5\,q[6]\times v[2]-k5\,q[2]\times v[5]-k5\,q[2]\times v[6])+\\
&\quad p[2]\,(k5\,q[5]\times v[2]+k5\,q[6]\times v[2]+k5\,q[2]\times v[5]+k5\,q[2]\times v[6])+\\
&\quad p[3]\,(-k1\,q[3]-k1\,q[4]+k1\,q[3]\times v[1]+k1\,q[4]\times v[1]+k1\,q[3]\times v[2]+k1\,q[4]\times v[2]+\\
&\qquad k1\,q[1]\times v[3]+k1\,q[2]\times v[3]+2\,k1\,q[3]\times v[3]+2\,k1\,q[4]\times v[3]+k1\,q[5]\times v[3]+\\
&\qquad k1\,q[6]\times v[3]+k1\,q[1]\times v[4]+k1\,q[2]\times v[4]+2\,k1\,q[3]\times v[4]+2\,k1\,q[4]\times v[4]+\\
&\qquad k1\,q[5]\times v[4]+k1\,q[6]\times v[4]+k1\,q[3]\times v[5]+k1\,q[4]\times v[5]+k1\,q[3]\times v[6]+k1\,q[4]\times v[6]))^2-\\
&(p[4]\,(-2\,k3\,q[1]\times q[3]-2\,k3\,q[1]\times q[4])+p[6]\,(-2\,k7\,q[1]\times q[5]-2\,k7\,q[1]\times q[6])+\\
&\quad p[1]\,(2\,k3\,q[1]\times q[3]+2\,k3\,q[1]\times q[4]+2\,k7\,q[1]\times q[5]+2\,k7\,q[1]\times q[6])+\\
&\quad p[5]\,(-2\,k5\,q[2]\times q[5]-2\,k5\,q[2]\times q[6])+p[2]\,(2\,k5\,q[2]\times q[5]+2\,k5\,q[2]\times q[6])+\\
&\quad p[3]\,(2\,k1\,q[1]\times q[3]+2\,k1\,q[2]\times q[3]+2\,k1\,q[3]^2+2\,k1\,q[1]\times q[4]+2\,k1\,q[2]\times q[4]+4\,k1\,q[3]\times q[4]+\\
&\qquad 2\,k1\,q[4]^2+2\,k1\,q[3]\times q[5]+2\,k1\,q[4]\times q[5]+2\,k1\,q[3]\times q[6]+2\,k1\,q[4]\times q[6]))\\
&(p[4]\,(-2\,k3\,v[1]\times v[3]-2\,k3\,v[1]\times v[4])+p[6]\,(-2\,k7\,v[1]\times v[5]-2\,k7\,v[1]\times v[6])+\\
&\quad p[1]\,(2\,k3\,v[1]\times v[3]+2\,k3\,v[1]\times v[4]+2\,k7\,v[1]\times v[5]+2\,k7\,v[1]\times v[6])+\\
&\quad p[5]\,(-2\,k5\,v[2]\times v[5]-2\,k5\,v[2]\times v[6])+p[2]\,(2\,k5\,v[2]\times v[5]+2\,k5\,v[2]\times v[6])+\\
&\quad p[3]\,(-2\,k1\,v[3]+2\,k1\,v[1]\times v[3]+2\,k1\,v[2]\times v[3]+2\,k1\,v[3]^2-\\
&\qquad 2\,k1\,v[4]+2\,k1\,v[1]\times v[4]+2\,k1\,v[2]\times v[4]+4\,k1\,v[3]\times v[4]+2\,k1\,v[4]^2+\\
&\qquad 2\,k1\,v[3]\times v[5]+2\,k1\,v[4]\times v[5]+2\,k1\,v[3]\times v[6]+2\,k1\,v[4]\times v[6]))
\end{aligned}
$$

Out[202]=

$$
\begin{aligned}
&\mathrm{xi}^2\,(p[4]\,(-2\,k3\,q[1]\times q[3]-2\,k3\,q[1]\times q[4])+p[6]\,(-2\,k7\,q[1]\times q[5]-2\,k7\,q[1]\times q[6])+\\
&\quad p[1]\,(2\,k3\,q[1]\times q[3]+2\,k3\,q[1]\times q[4]+2\,k7\,q[1]\times q[5]+2\,k7\,q[1]\times q[6])+\\
&\quad p[5]\,(-2\,k5\,q[2]\times q[5]-2\,k5\,q[2]\times q[6])+p[2]\,(2\,k5\,q[2]\times q[5]+2\,k5\,q[2]\times q[6])+\\
&\quad p[3]\,(2\,k1\,q[1]\times q[3]+2\,k1\,q[2]\times q[3]+2\,k1\,q[3]^2+2\,k1\,q[1]\times q[4]+2\,k1\,q[2]\times q[4]+4\,k1\,q[3]\times q[4]+\\
&\qquad 2\,k1\,q[4]^2+2\,k1\,q[3]\times q[5]+2\,k1\,q[4]\times q[5]+2\,k1\,q[3]\times q[6]+2\,k1\,q[4]\times q[6]))+\\
&2\,\mathrm{eta}\,\mathrm{xi}\,(p[4]\,(-k3\,q[3]\times v[1]-k3\,q[4]\times v[1]-k3\,q[1]\times v[3]-k3\,q[1]\times v[4])+\\
&\quad p[6]\,(-k7\,q[5]\times v[1]-k7\,q[6]\times v[1]-k7\,q[1]\times v[5]-k7\,q[1]\times v[6])+\\
&\quad p[1]\,(k3\,q[3]\times v[1]+k3\,q[4]\times v[1]+k7\,q[5]\times v[1]+k7\,q[6]\times v[1]+k3\,q[1]\times v[3]+k3\,q[1]\times v[4]+\\
&\qquad k7\,q[1]\times v[5]+k7\,q[1]\times v[6])+p[5]\,(-k5\,q[5]\times v[2]-k5\,q[6]\times v[2]-k5\,q[2]\times v[5]-k5\,q[2]\times v[6])+\\
&\quad p[2]\,(k5\,q[5]\times v[2]+k5\,q[6]\times v[2]+k5\,q[2]\times v[5]+k5\,q[2]\times v[6])+\\
&\quad p[3]\,(-k1\,q[3]-k1\,q[4]+k1\,q[3]\times v[1]+k1\,q[4]\times v[1]+k1\,q[3]\times v[2]+k1\,q[4]\times v[2]+\\
&\qquad k1\,q[1]\times v[3]+k1\,q[2]\times v[3]+2\,k1\,q[3]\times v[3]+2\,k1\,q[4]\times v[3]+k1\,q[5]\times v[3]+\\
&\qquad k1\,q[6]\times v[3]+k1\,q[1]\times v[4]+k1\,q[2]\times v[4]+2\,k1\,q[3]\times v[4]+2\,k1\,q[4]\times v[4]+\\
&\qquad k1\,q[5]\times v[4]+k1\,q[6]\times v[4]+k1\,q[3]\times v[5]+k1\,q[4]\times v[5]+k1\,q[3]\times v[6]+k1\,q[4]\times v[6]))+\\
&\mathrm{eta}^2\,(p[4]\,(-2\,k3\,v[1]\times v[3]-2\,k3\,v[1]\times v[4])+p[6]\,(-2\,k7\,v[1]\times v[5]-2\,k7\,v[1]\times v[6])+\\
&\quad p[1]\,(2\,k3\,v[1]\times v[3]+2\,k3\,v[1]\times v[4]+2\,k7\,v[1]\times v[5]+2\,k7\,v[1]\times v[6])+\\
&\quad p[5]\,(-2\,k5\,v[2]\times v[5]-2\,k5\,v[2]\times v[6])+p[2]\,(2\,k5\,v[2]\times v[5]+2\,k5\,v[2]\times v[6])+\\
&\quad p[3]\,(-2\,k1\,v[3]+2\,k1\,v[1]\times v[3]+2\,k1\,v[2]\times v[3]+2\,k1\,v[3]^2-\\
&\qquad 2\,k1\,v[4]+2\,k1\,v[1]\times v[4]+2\,k1\,v[2]\times v[4]+4\,k1\,v[3]\times v[4]+2\,k1\,v[4]^2+\\
&\qquad 2\,k1\,v[3]\times v[5]+2\,k1\,v[4]\times v[5]+2\,k1\,v[3]\times v[6]+2\,k1\,v[4]\times v[6]))
\end{aligned}
$$

In[203]:=

```
(* The model is quadratic in states and linear in ATOT, so all third
   derivatives with respect to the states vanish. *)
thirdStateTensor = Table[
 D[F[[i]], state[[j]], state[[k]], state[[ell]]],
 {i, 6}, {j, 6}, {k, 6}, {ell, 6}];
```

In[204]:=

```
Print["Fold normal form: aLP = p.B(Q,Q)/2 = ", aLP];
Print["Fold parameter coefficient: betaLP = p.dF/dATOT = ", betaLP];
Print["Fold curvature: kappaLP = -aLP/betaLP = ", kappaLP];
Print["Branching equation: ", branchingEquation, " = 0"];
Print["Branch discriminant: b^2-a c = ", branchDiscriminant];
Print["All third state derivatives vanish: ",
 Flatten[thirdStateTensor] === ConstantArray[0, 6^4]];

(* To evaluate numerical coefficients at a particular singularity, replace
   state, parameters, pNull, qNull, and v0 by the values returned by the
   exact labelling pipeline. The formulas use the manuscript's stated
   normalisation conventions. *)
```

Fold normal form: aLP = p.B(Q,Q)/2 =

$$\frac{1}{2}\,(p[4]\,(-2\,k3\,q[1]\times q[3]-2\,k3\,q[1]\times q[4])+p[6]\,(-2\,k7\,q[1]\times q[5]-2\,k7\,q[1]\times q[6])+$$
$$p[1]\,(2\,k3\,q[1]\times q[3]+2\,k3\,q[1]\times q[4]+2\,k7\,q[1]\times q[5]+2\,k7\,q[1]\times q[6])+$$
$$p[5]\,(-2\,k5\,q[2]\times q[5]-2\,k5\,q[2]\times q[6])+p[2]\,(2\,k5\,q[2]\times q[5]+2\,k5\,q[2]\times q[6])+$$
$$p[3]\,(2\,k1\,q[1]\times q[3]+2\,k1\,q[2]\times q[3]+2\,k1\,q[3]^2+2\,k1\,q[1]\times q[4]+2\,k1\,q[2]\times q[4]+4\,k1\,q[3]\times q[4]+$$
$$2\,k1\,q[4]^2+2\,k1\,q[3]\times q[5]+2\,k1\,q[4]\times q[5]+2\,k1\,q[3]\times q[6]+2\,k1\,q[4]\times q[6]))$$

Fold parameter coefficient: betaLP = p.dF/dATOT = $k1\,(KTOT-x3-x4)\,p[3]$

Fold curvature: kappaLP = −aLP/betaLP = $\dfrac{1}{2\,k1\,(KTOT-x3-x4)\,p[3]}$

$$(-p[4]\,(-2\,k3\,q[1]\times q[3]-2\,k3\,q[1]\times q[4])-p[6]\,(-2\,k7\,q[1]\times q[5]-2\,k7\,q[1]\times q[6])-$$
$$p[1]\,(2\,k3\,q[1]\times q[3]+2\,k3\,q[1]\times q[4]+2\,k7\,q[1]\times q[5]+2\,k7\,q[1]\times q[6])-$$
$$p[5]\,(-2\,k5\,q[2]\times q[5]-2\,k5\,q[2]\times q[6])-p[2]\,(2\,k5\,q[2]\times q[5]+2\,k5\,q[2]\times q[6])-$$
$$p[3]\,(2\,k1\,q[1]\times q[3]+2\,k1\,q[2]\times q[3]+2\,k1\,q[3]^2+2\,k1\,q[1]\times q[4]+2\,k1\,q[2]\times q[4]+$$
$$4\,k1\,q[3]\times q[4]+2\,k1\,q[4]^2+2\,k1\,q[3]\times q[5]+2\,k1\,q[4]\times q[5]+2\,k1\,q[3]\times q[6]+2\,k1\,q[4]\times q[6]))$$

Branching equation:

$$xi^2\,(p[4]\,(-2\,k3\,q[1]\times q[3]-2\,k3\,q[1]\times q[4])+p[6]\,(-2\,k7\,q[1]\times q[5]-2\,k7\,q[1]\times q[6])+$$
$$p[1]\,(2\,k3\,q[1]\times q[3]+2\,k3\,q[1]\times q[4]+2\,k7\,q[1]\times q[5]+2\,k7\,q[1]\times q[6])+$$
$$p[5]\,(-2\,k5\,q[2]\times q[5]-2\,k5\,q[2]\times q[6])+p[2]\,(2\,k5\,q[2]\times q[5]+2\,k5\,q[2]\times q[6])+$$
$$p[3]\,(2\,k1\,q[1]\times q[3]+2\,k1\,q[2]\times q[3]+2\,k1\,q[3]^2+2\,k1\,q[1]\times q[4]+2\,k1\,q[2]\times q[4]+4\,k1\,q[3]\times q[4]+$$
$$2\,k1\,q[4]^2+2\,k1\,q[3]\times q[5]+2\,k1\,q[4]\times q[5]+2\,k1\,q[3]\times q[6]+2\,k1\,q[4]\times q[6]))+$$
$$2\,eta\,xi\,(p[4]\,(-k3\,q[3]\times v[1]-k3\,q[4]\times v[1]-k3\,q[1]\times v[3]-k3\,q[1]\times v[4])+$$
$$p[6]\,(-k7\,q[5]\times v[1]-k7\,q[6]\times v[1]-k7\,q[1]\times v[5]-k7\,q[1]\times v[6])+$$
$$p[1]\,(k3\,q[3]\times v[1]+k3\,q[4]\times v[1]+k7\,q[5]\times v[1]+k7\,q[6]\times v[1]+k3\,q[1]\times v[3]+k3\,q[1]\times v[4]+$$
$$k7\,q[1]\times v[5]+k7\,q[1]\times v[6])+p[5]\,(-k5\,q[5]\times v[2]-k5\,q[6]\times v[2]-k5\,q[2]\times v[5]-k5\,q[2]\times v[6])+$$
$$p[2]\,(k5\,q[5]\times v[2]+k5\,q[6]\times v[2]+k5\,q[2]\times v[5]+k5\,q[2]\times v[6])+$$
$$p[3]\,(-k1\,q[3]-k1\,q[4]+k1\,q[3]\times v[1]+k1\,q[4]\times v[1]+k1\,q[3]\times v[2]+k1\,q[4]\times v[2]+$$
$$k1\,q[1]\times v[3]+k1\,q[2]\times v[3]+2\,k1\,q[3]\times v[3]+2\,k1\,q[4]\times v[3]+k1\,q[5]\times v[3]+$$
$$k1\,q[6]\times v[3]+k1\,q[1]\times v[4]+k1\,q[2]\times v[4]+2\,k1\,q[3]\times v[4]+2\,k1\,q[4]\times v[4]+$$
$$k1\,q[5]\times v[4]+k1\,q[6]\times v[4]+k1\,q[3]\times v[5]+k1\,q[4]\times v[5]+k1\,q[3]\times v[6]+k1\,q[4]\times v[6]))+$$
$$eta^2\,(p[4]\,(-2\,k3\,v[1]\times v[3]-2\,k3\,v[1]\times v[4])+p[6]\,(-2\,k7\,v[1]\times v[5]-2\,k7\,v[1]\times v[6])+$$
$$p[1]\,(2\,k3\,v[1]\times v[3]+2\,k3\,v[1]\times v[4]+2\,k7\,v[1]\times v[5]+2\,k7\,v[1]\times v[6])+$$
$$p[5]\,(-2\,k5\,v[2]\times v[5]-2\,k5\,v[2]\times v[6])+p[2]\,(2\,k5\,v[2]\times v[5]+2\,k5\,v[2]\times v[6])+$$
$$p[3]\,(-2\,k1\,v[3]+2\,k1\,v[1]\times v[3]+2\,k1\,v[2]\times v[3]+2\,k1\,v[3]^2-2\,k1\,v[4]+$$
$$2\,k1\,v[1]\times v[4]+2\,k1\,v[2]\times v[4]+4\,k1\,v[3]\times v[4]+2\,k1\,v[4]^2+$$
$$2\,k1\,v[3]\times v[5]+2\,k1\,v[4]\times v[5]+2\,k1\,v[3]\times v[6]+2\,k1\,v[4]\times v[6]))=0$$

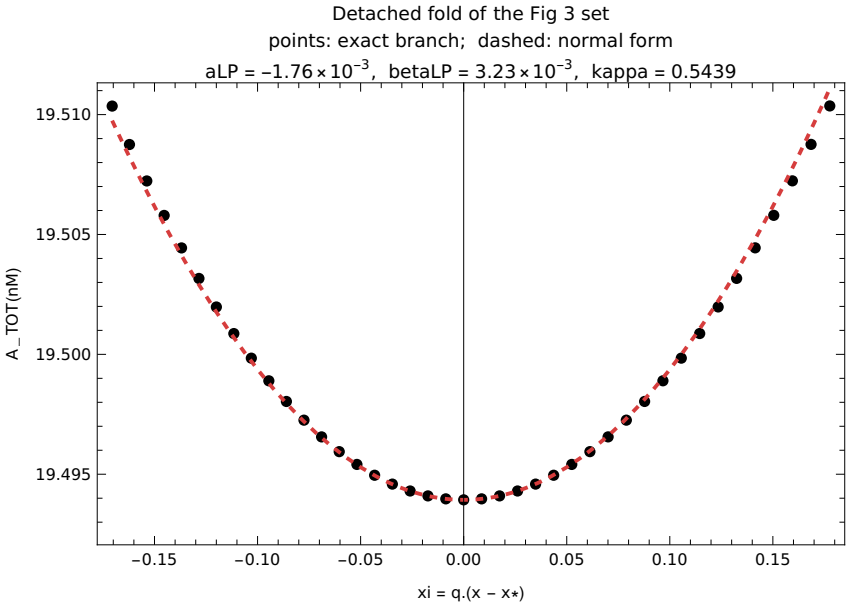
```
Branch discriminant: b^2-a c =
 (p[4] (-k3 q[3] × v[1] - k3 q[4] × v[1] - k3 q[1] × v[3] - k3 q[1] × v[4]) + p[6]
       (-k7 q[5] × v[1] - k7 q[6] × v[1] - k7 q[1] × v[5] - k7 q[1] × v[6]) + p[1] (k3 q[3] × v[1] + k3 q[4] × v[1] +
          k7 q[5] × v[1] + k7 q[6] × v[1] + k3 q[1] × v[3] + k3 q[1] × v[4] + k7 q[1] × v[5] + k7 q[1] × v[6]) +
      p[5] (-k5 q[5] × v[2] - k5 q[6] × v[2] - k5 q[2] × v[5] - k5 q[2] × v[6]) +
      p[2] (k5 q[5] × v[2] + k5 q[6] × v[2] + k5 q[2] × v[5] + k5 q[2] × v[6]) +
      p[3] (-k1 q[3] - k1 q[4] + k1 q[3] × v[1] + k1 q[4] × v[1] + k1 q[3] × v[2] + k1 q[4] × v[2] +
          k1 q[1] × v[3] + k1 q[2] × v[3] + 2 k1 q[3] × v[3] + 2 k1 q[4] × v[3] + k1 q[5] × v[3] +
          k1 q[6] × v[3] + k1 q[1] × v[4] + k1 q[2] × v[4] + 2 k1 q[3] × v[4] + 2 k1 q[4] × v[4] +
          k1 q[5] × v[4] + k1 q[6] × v[4] + k1 q[3] × v[5] + k1 q[4] × v[5] + k1 q[3] × v[6] + k1 q[4] × v[6])^2 -
   (p[4] (-2 k3 q[1] × q[3] - 2 k3 q[1] × q[4]) + p[6] (-2 k7 q[1] × q[5] - 2 k7 q[1] × q[6]) +
      p[1] (2 k3 q[1] × q[3] + 2 k3 q[1] × q[4] + 2 k7 q[1] × q[5] + 2 k7 q[1] × q[6]) +
      p[5] (-2 k5 q[2] × q[5] - 2 k5 q[2] × q[6]) + p[2] (2 k5 q[2] × q[5] + 2 k5 q[2] × q[6]) +
      p[3] (2 k1 q[1] × q[3] + 2 k1 q[2] × q[3] + 2 k1 q[3]^2 + 2 k1 q[1] × q[4] + 2 k1 q[2] × q[4] + 4 k1 q[3] × q[4] +
          2 k1 q[4]^2 + 2 k1 q[3] × q[5] + 2 k1 q[4] × q[5] + 2 k1 q[3] × q[6] + 2 k1 q[4] × q[6])) *
   (p[4] (-2 k3 v[1] × v[3] - 2 k3 v[1] × v[4]) + p[6] (-2 k7 v[1] × v[5] - 2 k7 v[1] × v[6]) +
      p[1] (2 k3 v[1] × v[3] + 2 k3 v[1] × v[4] + 2 k7 v[1] × v[5] + 2 k7 v[1] × v[6]) +
      p[5] (-2 k5 v[2] × v[5] - 2 k5 v[2] × v[6]) + p[2] (2 k5 v[2] × v[5] + 2 k5 v[2] × v[6]) +
      p[3] (-2 k1 v[3] + 2 k1 v[1] × v[3] + 2 k1 v[2] × v[3] + 2 k1 v[3]^2 - 2 k1 v[4] +
          2 k1 v[1] × v[4] + 2 k1 v[2] × v[4] + 4 k1 v[3] × v[4] + 2 k1 v[4]^2 +
          2 k1 v[3] × v[5] + 2 k1 v[4] × v[5] + 2 k1 v[3] × v[6] + 2 k1 v[4] × v[6]))

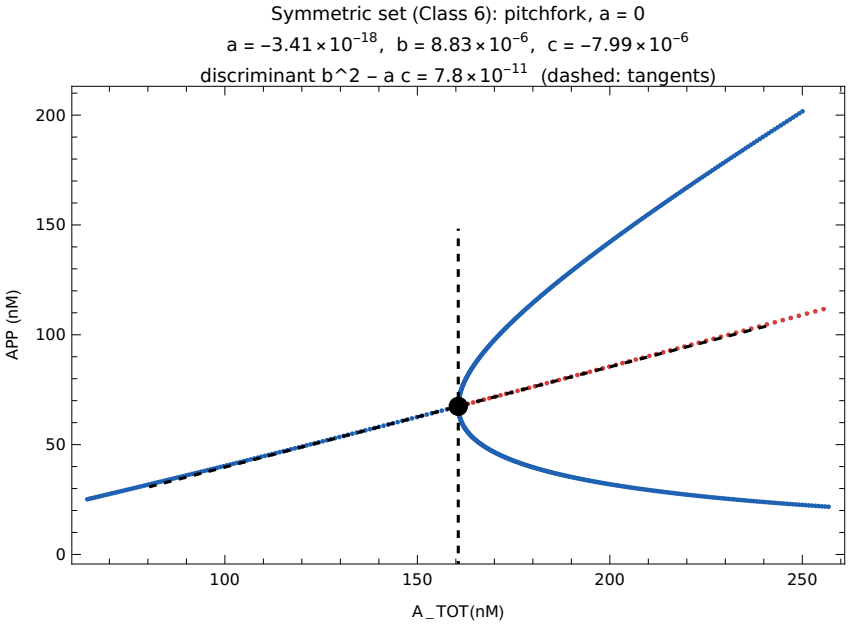
All third state derivatives vanish: True
```

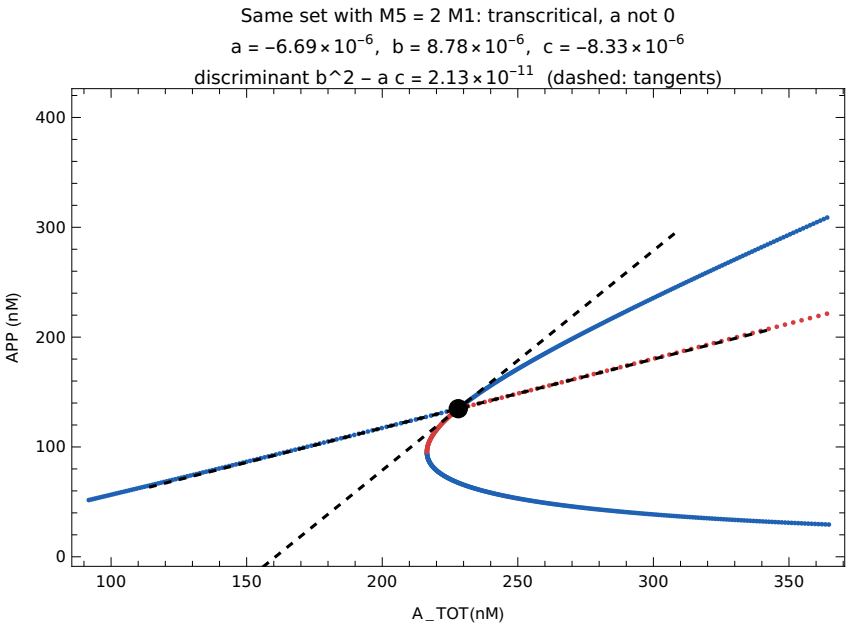
Added figure. Plots: the coefficients of Section 9 evaluated numerically. (Left) at the detached fold of the Fig 3 set, A_TOT along the exact branch against xi = q.(x - x*) (points) and the normal form A_TOT* + kappa xi^2 with kappa = -aLP/betaLP (dashed). (Middle, right) at the branch point of the symmetric set (pitchfork, a = 0) and of the same set with M5 = 2 M1 (transcritical, a not 0): exact branches near the branch point and the two tangent directions of the branching equation a xi^2 + 2 b xi eta + c eta^2 = 0 (dashed). Saved as S9_normal_forms.pdf.

In[210]:=

```
(* Plot for Section 9: fold and branch-point normal-form coefficients, evaluated numeric
Module[{pr = figSet3, c = figConst[figSet3], fs, xs, at, jm, q, p, hm, qq, aLP, bLP, kap,
 fs = figDetachedFold[pr]; xs = fs["x"]; at = fs["ATOT"];
 jm = figJac[pr, at, xs]; {q, p} = Most[figNull[jm]];
 hm = figHess[pr, at, xs]; qq = Append[q, 0];
 aLP = (p . (qq . # . qq & /@ hm))/2; bLP = p . figFA[pr, at, xs]; kap = -aLP/bLP;
 pts = Table[With[{s = figState[pr, c, fs["u"] 10^dd, figA[c, fs["u"] 10^dd]]}, {q . (s["x"] -
 gF = Show[ListPlot[pts, PlotStyle -> Directive[Black, PointSize[0.015]]],
   Plot[at + kap figXi^2, {figXi, Min[pts[[All, 1]]], Max[pts[[All, 1]]]}, PlotStyle -> Direct
   Frame -> True, FrameLabel -> {"xi = q.(x - x*)", "A_TOT (nM)"}, ImageSize -> 440,
   PlotLabel -> Style[Column[{"Detached fold of the Fig 3 set", "points: exact branch;
      Row[{"aLP = ", ScientificForm[aLP, 3], ",  betaLP = ", ScientificForm[bLP, 3], ",  k
 gP = figBPPlot[figSet6, "Symmetric set (Class 6): pitchfork, a = 0"];
 trans = Module[{qs = figSet6}, qs["k5"] = figSet6["k5"]/2; qs];
 gT = figBPPlot[trans, "Same set with M5 = 2 M1: transcritical, a not 0"];
 figShow["S9_normal_forms",{gF, gP, gT}, GraphicsRow[{gF, gP, gT}, ImageSize -> 1300]]]
```


Detached fold of the Fig 3 set
points: exact branch; dashed: normal form
aLP = −1.76 × 10^−3, betaLP = 3.23 × 10^−3, kappa = 0.5439
19.510
19.505
19.500
19.495
A_TOT(nM)
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
xi = q.(x − x*)


Symmetric set (Class 6): pitchfork, a = 0
a = −3.41 × 10^−18, b = 8.83 × 10^−6, c = −7.99 × 10^−6
discriminant b^2 − a c = 7.8 × 10^−11 (dashed: tangents)
200
150
100
50
0
APP (nM)
100
150
200
250
A_TOT(nM)



saved:
/wolframcloud/userfiles/9ff/9ffe219c-88d9-4035-be0c-26993d670000/Files/Harika/Derivation_Notebooks/Final/plots/S9_normal_forms.pdf